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Slosh Damping with Floating Magnetoactive Micro-Baffles

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SLOSH DAMPING WITH FLOATING MAGNETOACTIVE MICRO-BAFFLES

by

VIJAY SANTHANAM

A Thesis Submitted to the College of Engineering, Department of Aerospace Engineering
for the partial fulfillment of the requirements of the degree of

Master of Science in Aerospace Engineering

Embry-Riddle Aeronautical University

Daytona Beach, Florida

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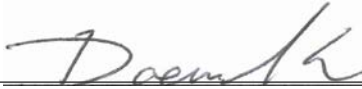
SLOSH DAMPING WITH FLOATING MAGNETOACTIVE
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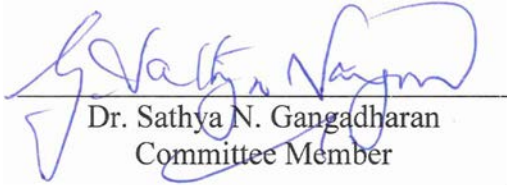
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This thesis was prepared under the direction of the candidate's Thesis Committee Chair, Dr. Daewon Kim, Assistant Professor, Daytona Beach Campus, and Thesis Committee Members Dr. Sathya N. Gangadharan, Professor, Daytona Beach Campus, and Dr. Mark A. Ricklick, Assistant Professor, Daytona Beach Campus, and has been approved by the Thesis Committee. It was submitted to the Department of Aerospace Engineering in partial fulfillment of the requirements for the degree of Master of Science in Aerospace Engineering

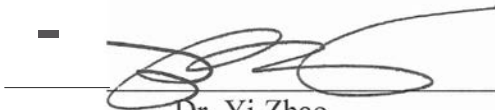
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ABSTRACT

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Liquid sloshing within propellant tanks of launch vehicles and other major vehicles has been a major concern. Various methods have been utilized for the damping of slosh through Propellant Management Devices (PMD) accomplishing a wide range of results. Exploratory research conducted at the Embry-Riddle Aeronautical University Fuel Slosh Test Facility in development of an innovative PMD is presented. Embedding floating micro-baffles with a magnetoactive material such that the baffle can be manipulated when exposed to a magnetic field preserves the benefits of both floating and static baffle designs. Activated micro-baffles form a rigid layer at the free surface and provide a restriction of the fluid motion. Proposed micro-baffle design and magnetic activation source method along with proof-of-concept experiments comparing the scope of this research to previous PMD methods are presented. A computational fluid dynamics approach is outlined to compliment these experimental results.

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1. INTRODUCTION

1.1 Sloshing

The presence of liquid in a tank which is exposed to dynamic conditions has been under review for quite some time. Such scenarios may occur in liquid propellant rockets, aircraft propellant tanks, ships, petroleum tankers, and other applications. In engineering, the phenomenon of sloshing is one of great importance due to its existence in numerous scenarios consisting of liquids from the common teacup to liquid carrying ships allowing them to share a common feature of unrestrained liquid free surfaces. Sloshing can cause the tank system to deviate due to the buildup of kinetic energy of the liquid and its consequent interaction to the walls of the container [1]. Figure 1 shows a display of the occurrence of slosh in a spherical container when agitated.

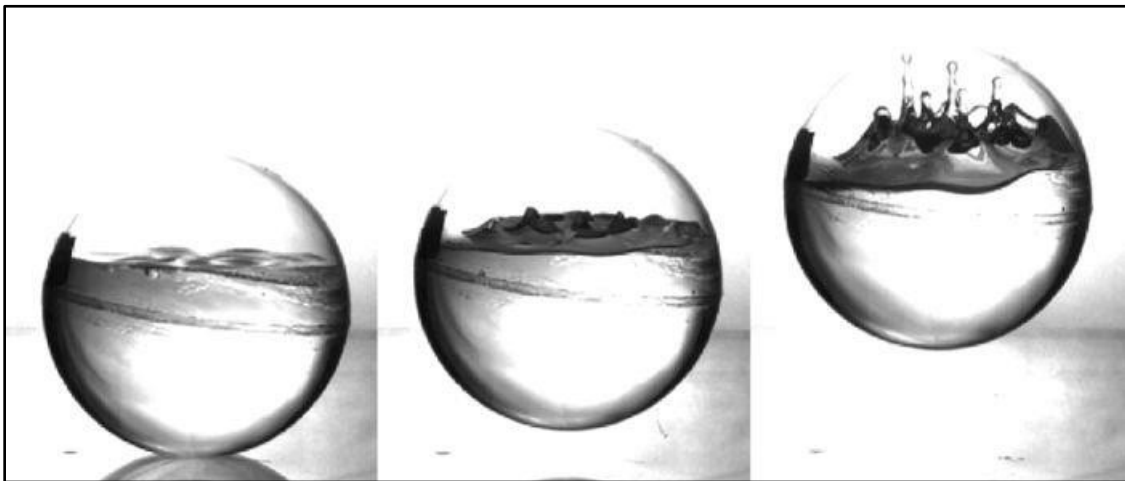


Figure 1: Sloshing in a spherical container [17]

Fluid sloshing is defined as the periodic motion of fluid with the free surface in a liquid container. The hydrodynamic forces exerted due to sloshing pose a risk to the structural integrity of the tank walls [2]. These forces lead to the occurrence of a phenomenon that constitutes the focal point of interest in the motions of these containers and has remained a challenging problem in the field of mechanics. This concern is augmented with the constantly increasing size of space vehicles and rocket vehicle propellant containers and the significant dynamic forces they are exposed to [3].



Figure 2: Sloshing occurring in a tanker [18]

The resolution of sloshing is, therefore, of significant practical importance to many engineering disciplines, such as aerospace, civil, mechanical and marine. In aerospace technology, sloshing is a well-recognized problem, particularly in liquid propellant launch vehicles, which have an enormous percentage of their initial weight as fuel. When slosh waves are allowed to freely oscillate, they have a tendency to reach resonance. At resonance, slosh waves have maximum amplitude. The forces of sloshing propellant cause the spacecraft to rotate about its spin axis. Traditionally, thrust vector correction methods are used to correct the nutation in the spacecraft. However, the high magnitudes of propellant sloshing forces overpower the corrections being made. This results in an increase in nutation and complete loss of the control of the spacecraft. This is seen in the 1969 NASA's ATS V mission failed due to nutation growth caused by liquid propellant sloshing [4]. Mission failures caused due to unpredictable behavior of propellant slosh call for an in-depth and detailed study of propellant slosh behavior and parameter estimation.

1.2 Anti-Sloshing and Propellant Management Devices

Anti-sloshing is the process of damping the remnant slosh waves from their resonant frequency condition. This reduces the force acting on the wall of the container in which the fluid exists. Historically, propellant slosh can be predicted accurately by various numeric models developed during 1960s. The difficulty in prediction increases with addition of passive Propellant Management Devices (PMDs) like diaphragms and baffles to the tanks. Passive PMDs work for limited range of propellant slosh frequency, they

take up space in the fuel tank and are heavy and bulky. Thus, weight to operation range ratio of passive PMDs is poor. On the other hand, active damping mechanism would take up some space of the fuel tank, but the versatile frequency operation of these devices makes them more useful and advantageous than passive PMDs. Through the years, various techniques to dampen the motion of the liquid propellant have come into existence and such techniques involve passive structural devices and active mechanisms [5]. A diaphragm is a type of passive PMD that is implemented within the propellant tank itself. As the liquid propellant leaves the tank, the diaphragm compresses and lessens the amount of surface area in which the liquid would originally slosh. Figure 3, below shows the classification.

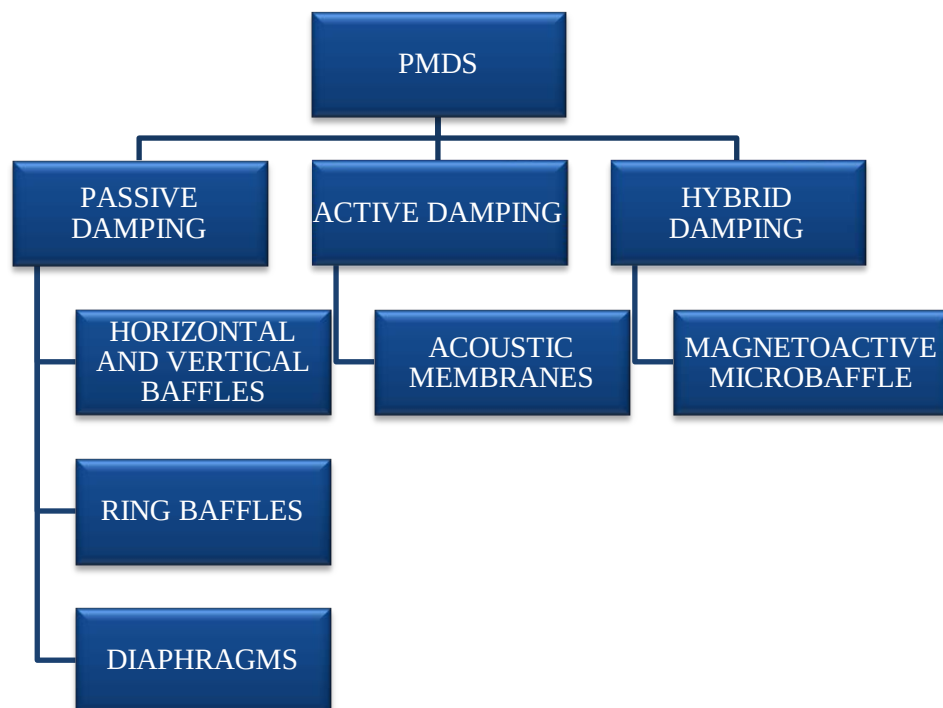


Figure 3: Classification of PMDs

A baffle is a similar type of passive PMD that creates a simple barrier that acts to restrict the physical motion of the liquid thus damping the amount of slosh it can exhibit. The two main types of baffles are for wall-fixed and for floating types; each accompanied by their own set of advantages and disadvantages [3]. The primary advantage of floating baffles is the reduced weight over their wall-fixed counterparts. Additionally, wall-fixed baffles are limited in their placement with respect to the tank walls due to the presence of supporting structures which may be located on the tank walls.

Active damping mechanisms physically create waves that destructively interfere with the liquid propellant undergoing sloshing. By changing the frequency at which these waves are created, the magnitude of the sloshing waves can be controlled. Should the operating frequency of the mechanism coincide with the resonant frequency of the propellant, the sloshing waves of the propellant would be completely dampened instantaneously. Unlike the passive PMDs, this mechanism has never been implemented into an operating space system. Some of the designs include mounting baffles of various shapes to the inner sidewall of the tank [6].

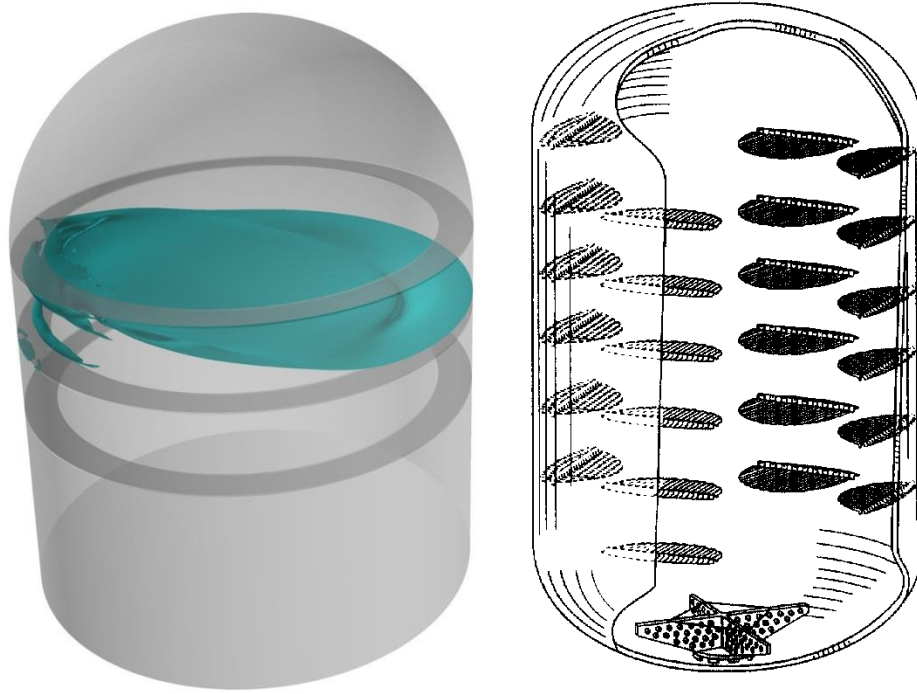


Figure 4: Tanks incorporating ring baffles (left) and horizontal baffles (right)

1.3 Past efforts in Anti-Sloshing

Before the advances in Computational Fluid Dynamics (CFD) and computing power, propellant anti-slosh issue was dealt on a case by case basis. Experimental setup would be designed to validate mathematically predicted parameters such as frequency and damping for propellant slosh. This process was time consuming and economically draining. The parameters thus obtained were used to design passive PMD like baffles and diaphragms for propellant slosh damping. These passive damping devices provide excellent propellant slosh damping for a small range of frequency and small amplitude of sloshing, but they are not effective when propellant fill level changes and sloshing

frequency is outside their design range. These devices are also bulky, consume space, add significant weight and have small operation range.

Previous research of slosh suppression via floating baffles has indicated that successful damping of fluid slosh is carried out when baffle members interact by colliding with one another thus absorbing the kinetic energy imparted to the fuel upon movement of the tank [7]. Other baffle methods are defined as successful by increasing the natural frequency of the tank sections and decreasing the wave amplitude at the free surface. A study on the analysis of boundary element models for liquid sloshing concludes that the effect of the baffle position on the slosh reduction is more important than the geometry and location of the baffle especially if the baffles are appropriately located near the free surface [2].

Presently active damping devices are being developed to overcome the disadvantages of passive damping devices. Active damping devices work for a wide range of amplitude and frequencies and for all the propellant fill level in the tanks. An active damping mechanism consists of a device that can generate high frequency small amplitude waves with opposite phase to that of sloshing waves.

With advances in CFD algorithms, computing power and visual graphics on computers, CFD solvers have introduced new methods for simulation of engineering problems. CFD model once set can be manipulated to simulate wide range of scenarios with complex conditions applied on the model to understand physical behavior of the actual prototype without wasting time and money on designing new experimental setups. Conditions like absolute zero gravity can be simulated in CFD software along with varying geometry and

other parameters. Due to such versatile application, CFD software is being used on large scale in aerospace industries for simulation purpose. They provide a means of experimental data without actually performing the experiment.

Sloshing phenomenon is affected by gravitational forces, surface tension and surface kinematics. Accounting for all detail dynamics for complete vehicle dynamic modeling is time consuming. Sloshing dynamics can be represented by spring mass damping mechanical model. Embry Riddle Aeronautical University (ERAU) has developed a parameter estimation process using MATLAB SimMechanics software [1, 8] for propellant slosh. Propellant slosh at free surface is analogous to mechanical pendulum inside a tank. For the parameter estimation, a dynamic model of fuel slosh inside a laterally oscillating tank is used. The propellant inside the tank can be divided and modeled into two parts. First part of the propellant is stationary with respect to the tank and can be modeled as fixed mass at the bottom of the tank. The second part of the propellant sloshes and can be modeled as a small mass pendulum hanging from a pivot point by a mass-less string with a spring and damper combination that would account for viscous forces. Experimental data is used in combination with MATLAB SimMechanics model to extract parameters including pendulum mass, damping, pivot position, fixed mass and its location.

ERAU has also performed experimental and CFD research on passive PMD like diaphragm [9]. For passive PMD research, ANSYS FLUENT CFD software coupled with Fluid Structure Interaction (FSI) was used for modeling and solving propellant slosh.

This research confirmed capability of ANSYS CFD software to solve CFD-FSI models and parameterize propellant slosh dynamics. New capability of CFD solvers of solving multiphase simulations has made modeling of complex propellant slosh problem possible in this software. With multiphase algorithms being included, modeling of multiple fluids like air and water with a clear boundary separating both the fluids is possible now.

1.4 Floating Magnetoactive Micro-baffles

Previous research of slosh suppression via floating baffles has indicated that successful damping of fluid slosh is based on the same principle; baffle members interact by colliding with one another thus absorbing the kinetic energy imparted to the fuel upon movement of the tank [6]. Other baffle methods are defined as successful by increasing the natural frequency of the tank sections and decreasing the wave amplitude at the free surface. A study on the analysis of boundary element models for liquid sloshing concludes that the effect of the baffle position on the slosh reduction is more important than the geometry and location of the baffle especially if the baffles are appropriately located near the free surface.

The concept of floating magnetoactive micro-baffles is adapted from previous research involving the constrained floating baffle, and also the use of a diaphragm on the free surface [6, 10]. It provides a structural layer to constrain the free surface of the liquid. Floating micro-baffles allow the damping of slosh at all filled levels of the tank and also adapt to the attitude of the vehicle. Upon exposure to magnetic fields of varying strength,

it is predicted that the rigidity of the structural layer can adapt to various slosh conditions. Furthermore, this slosh suppression method preserves the natural frequency increase and kinetic energy interactions that correspond to both wall-fixed and floating baffle PMDs. The challenges encountered and the milestones achieved at this stage of the development of floating magnetoactive micro-baffles; specifically in the development of the micro-baffle design, the magnetic activation source, and proof-of-concept experiments are documented.

2. METHOD OF APPROACH

The overall goal of this research is to create a semi rigid structural grid at the free surface of the liquid upon activation of the floating magnetoactive micro-baffles. The primary objective in achieving this goal is to determine a suitable activation method. This may include, but is not limited to, orientation and strength of the magnetic source which activates the micro-baffles as desired. The secondary objective is to define parameters which characterize the rigidity of the structural grid; namely the strength between magnetically bonded elements. At this point in the research, efforts towards the primary and secondary objectives have primarily focused on electromagnet design/assembly and modifications to the micro-baffle geometry and magnetoactive substance.

Based on these objectives, the secondary goal of this research is to provide a conceptual validation of using floating magnetoactive micro-baffles for the purpose of anti-sloshing by experimental analysis and Computational Fluid Dynamics (CFD) analysis. The effectiveness of the magnetoactive micro-baffles are tested in the slosh tank facility at ERAU. The computational approach provides a comparative simulation of the slosh experiment. The experimental approach of this problem provides a realistic visualization of the problem and helps to define the modeling after using CFD.

3. EXPERIMENTAL APPROACH

3.1 Experimental Setup

The experimental setup in the figure below comprises of a laterally exciting platform on which a spherical or cylindrical tank is mounted. Several research efforts in this platform have been carried out over a period of time.

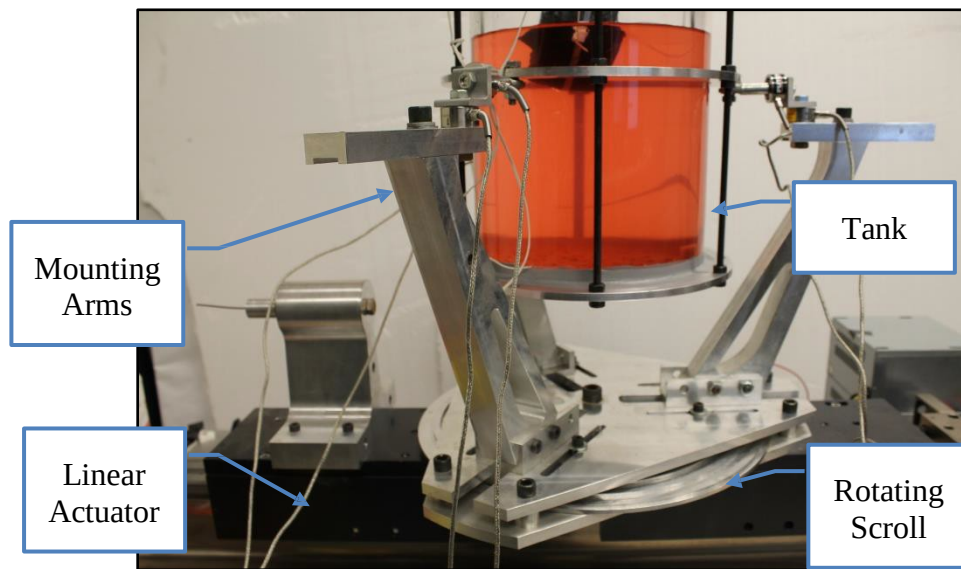


Figure 5: Slosh test bed holding a cylindrical tank

The current experimental setup portrayed in Figure 5 consists of a dynamic force-balance, fixed to an Aerotech single axis Linear Motion Actuator (LMA). Sloshing is generated in a lateral direction by providing a mechanical frequency of given amplitude and the tank is held by arms which are equipped by Futek dynamic load cells to measure the forces acting on the walls of the tank.

The setup is equipped with an adjustable rotary scroll which allows the setup to accommodate a variety of tank sizes and shapes ranging from 8" to 16" in diameter [9].

The tank used in this research is an 8" cylindrical tank.

The actuation is programmed by Aerotech's Soloist CP Software, which receives control from a unique LabVIEW Code developed at ERAU. Data from experimental simulations is acquired via these six dynamic load cells located around the centerline of the experimental tank. These load cells resolve the forces and moments on the sidewalls of the tank into the radial, tangential and vertical directions. For the purpose of proof of concept, only the load cell sensing the maximum reaction forces acting along the motion direction is used.

3.2 Process of Experimental Analysis

The figure below shows a flowchart of the general process of the experimental approach.

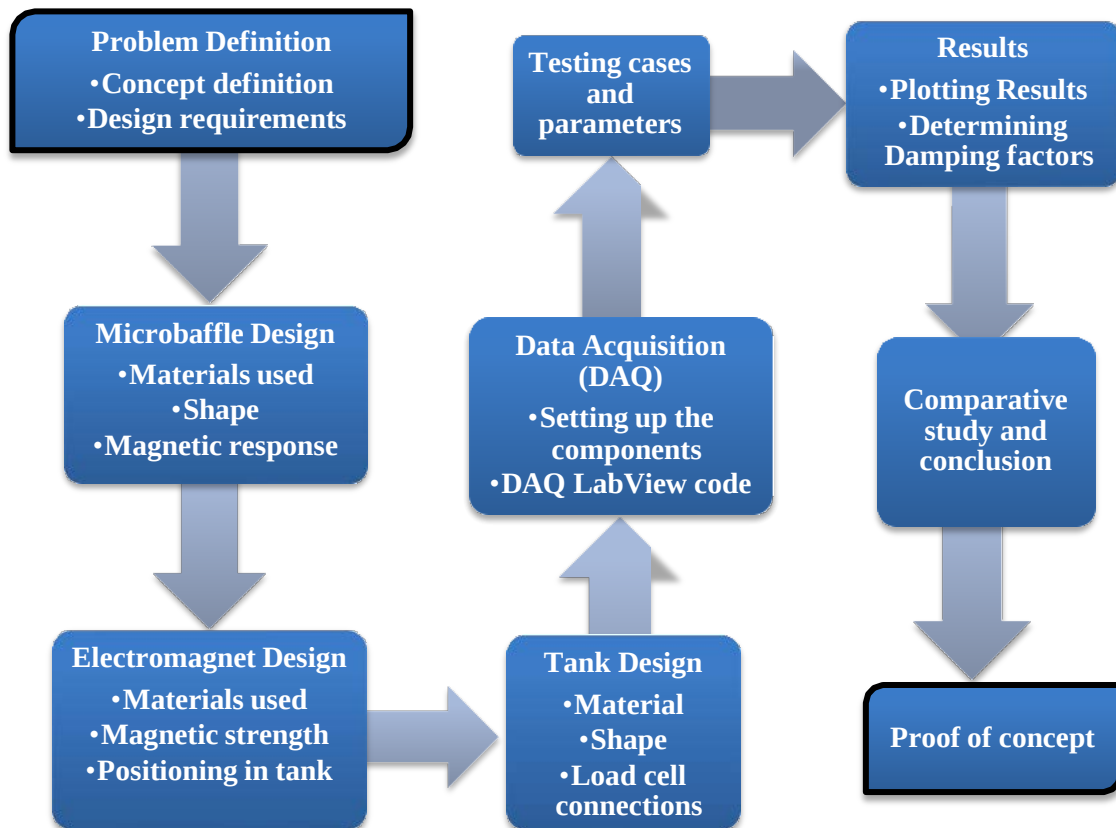


Figure 6: Process of experimental analysis

3.3 Design and construction of the micro-baffle

Extensive iterations in the geometry of the micro-baffle design and magnetoactive agent have been conducted. Modification of the type and amount of magnetoactive material embedded within the chamber has a direct effect on the required buoyancy to keep the micro-baffle afloat. A CATIA model of the most current magnetoactive micro-baffle revision is illustrated in Figure 7.

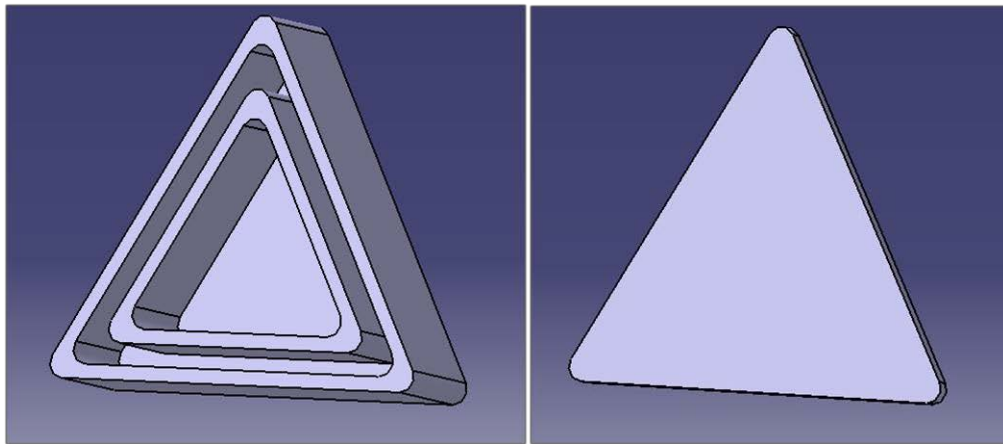


Figure 7: Micro-baffle base (left) and lid (right) 3D models

The current design is a triangular base-lid shape printed from Acrylonitrile Butadiene Styrene (ABS) plastic and Polylactic Acid (PLA) plastic with 1.5" long sides and 0.4" thick with an inner structure of two components; an air chamber for added buoyancy and a channel 0.075" wide along the perimeter to house the magnetoactive material. While other cross sectional shapes had been considered for the experiment, an array of equilateral triangles showed to be the most viable option for this particular experiment within a circular cross sectional area of the tank, which results in maximum coverage of

the free surface area. ABS and PLA plastics are both relatively light and buoyant, which are important parameters for this experiment [11]. Upon locating the magnetoactive material into the inner channel, the lid is secured using PVC cement.

Initially, micro-baffles are filled with various mixture solutions of Carbonyl Iron (CI) particles and ferrofluid. As research efforts progressed and more is learned about the interaction between adjacent activated micro-baffles and the magnetic source, it is concluded that the magnetoactive material embedded within the micro-baffle structure needs to be one that preserves the magnetic flux which is transmitted through the micro-baffle from the magnet onto the adjacent micro-baffle. That is, a material that is as magnetically permeable as possible. This conclusion is realized when two adjacent micro-baffles filled with just CI particles formed a stronger bond than adjacent micro-baffles filled with just ferrofluid. Adjacent micro-baffles embedded solely with CI particles are capable of withstanding an inertial load as opposed to their ferrofluid counterparts. Magnetic permeability is defined as a measure of the ability of a substance to sustain a magnetic field [12]. With the exception of several man-made alloys and other magnetic mediums, raw iron offers the highest permeability value. The table below lists relative permeability values for a variety of materials [13].

Table 1: Relative permeability of materials

Material	Relative Permeability	Material	Relative Permeability
Wood	~1.0	Aluminum	~1.0
Stainless Steel (<i>martensitic, hardened</i>)	40-95	Stainless Steel (<i>martensitic, annealed</i>)	750-950
Iron (<i>various</i>)	5,000-200,000	Metglas	1,000,000

At this point in the research, steel nails offer a cost-effective and readily available solution and are being used as the magnetoactive material located within the micro-baffle channels. Hence, based on the relative permeability values, to provide an even stronger bond the magnetoactive material chosen are steel nails that are embedded into the channels of the micro-baffle as shown in Figure 8.

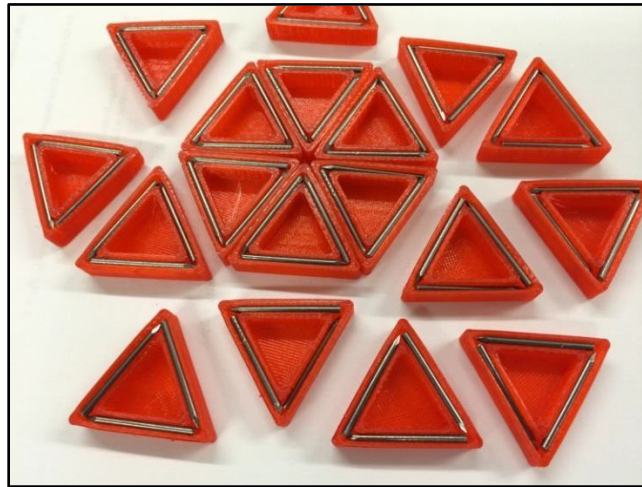


Figure 8: 3D Printed baffles (open) with embedded steel nails

3.4 Design and Construction of the Magnetic Activation Source

Throughout the course of the research, modified solenoids, electromagnets assembled in the ERAU Fuel Slosh Test Facility, and bar magnets have been used to assess necessary requirements for the magnetic activation source. The source must be designed such that the flux lines emanate far from the pole of the magnet through the diameter of the tank. Therefore, a simple test setup is devised to move the magnetic source in proximity of an magnetoactive micro-baffle floating in a cup of water to quantify the “influence range”. The governing equation for the pole strength, in Tesla, of an electromagnet is equal to:

$$P = \frac{NAI}{L} \quad (1)$$

Where N is the number of turns, I is the current, and A is the cross-sectional area of the core and L is the length of the core [14].

When initial electromagnet assembly attempts and solenoid purchases proved unsuccessful in obtaining a suitable magnetic source, Equation (1) is used in conjunction with the results of the influence range experiment to build a strong electromagnet.

This electromagnet is the current electromagnet used for all slosh tests and is made from 600 ft. of 18 gauge wire, and a 0.75" steel pipe fitting core, 1.5" long. Conduit plates are used as end caps and a 0.75" thick steel bolt is filed to fit into the pipe fitting after the wire is wound. The 18 gauge wire used for the electromagnet has a cross sectional area of 0.823 square inches and maximum current of 2.3 amperes [15]. Compared to the 6.5 cm

influence range of the most effective solenoid, the current electromagnet has an influence range of ~ 20.0 cm.



Figure 9: Electromagnet used for the activation source

The electromagnet is then spray coated using a Flex-Seal sealant. This is to allow the electromagnet to resist water from entering inside. The magnet is then hung from the top of the tank and half immersed into the water. The height is set such that the axis of the core lies along the level of the free surface and the axis direction is along the axis of the linear motion of the actuator.

3.5 Selection of Experimental Tank and Design

The figure below shows the fabricated cylindrical tank that is used for the experimental setup. It is made using an 8" diameter 0.25" cylindrical clear acrylic tube of 12" height. It is reinforced by 0.25" thick aluminum steel rings to provide the mounts to attach to the test rig.



Figure 10: Cylindrical tank used for the experiment

3.6 Experimental Test Bed Setup

The tank as shown in Figure 11 is filled with propellant 60% by volume, which has been identified through experimental testing to induce maximum slosh amplitude. The testing fluid is water instead of common rocket propellant, hydrazine, because they are physically very similar. The tank is open at the top for ease of access to place the magnetic activation source and micro-baffles.

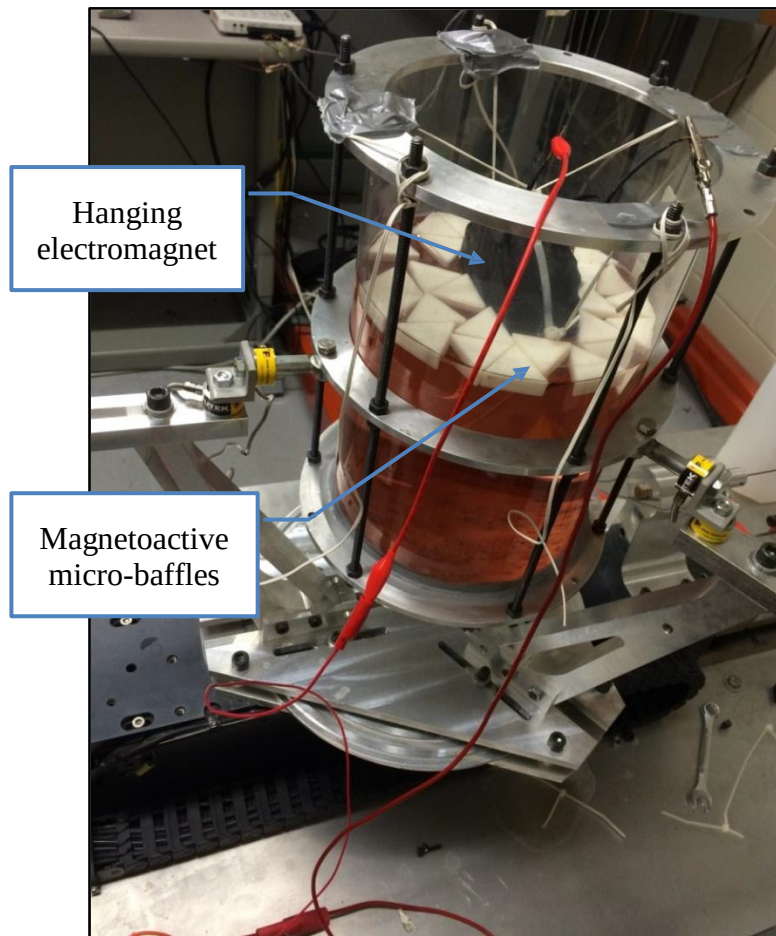


Figure 11: Slosh Test Bed setup with the magnetic source mounted

Load cell output signals are refined using a low pass filter (LPF) incorporated into a LabVIEW code. The damping effect is characterized by evaluating the signal immediately after the point in time where actuation of the Aerotech system has ceased. This is known as the remnant slosh period and the damping coefficient of this signal corresponds to the amount of slosh suppression present. The force signals and damping coefficients between each of the testing conditions are compared.

3.7 Data Acquisition

The National Instruments myDAQ is used in conjunction with a custom built signal conditioner to acquire the data signals from the load cells and is filtered by a LabVIEW code.

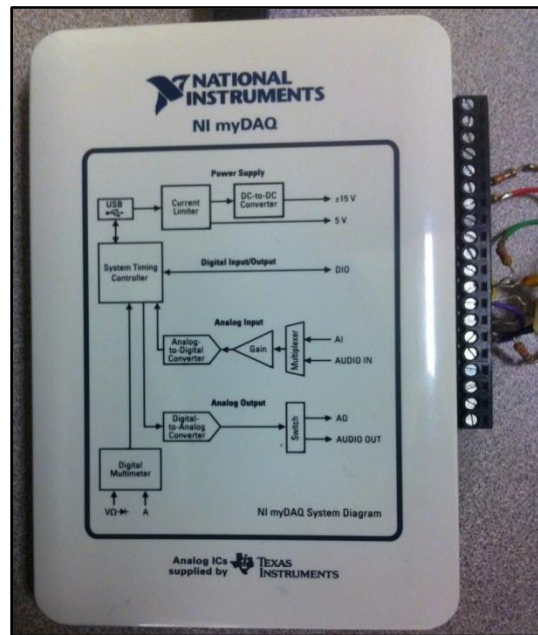


Figure 12: The National Instruments MyDAQ

The LabVIEW code was developed at the ERAU Slosh Test Facility. It is a custom data acquisition code that also provides a control interface for the linear actuator. The code is designed to read the user input parameters for actuator stage displacement, frequency and number of cycles. The linear actuator in turn operates according to these input parameters. The load cells collect the force data acting on the tank wall and the signals from these load cells are passed through a signal generator and then through the Data

The screenshot displays a Simulink control system model for a linear actuator. The model is titled "2 [0..3]". It features an "Init" block, a "Simulation Run Time (ms)" block, and a "Linear Actuator Position (mm)" block. The system is controlled by a feedback loop with a gain of 5.64. The output is compared with a reference value of 1.38. The model includes two "Isklat Siyvalar" (Signal Out) blocks, each with a "Signal Out" port. The output of the first block is compared with the reference value 1.38, and the output of the second block is compared with the reference value 3.8. The model also includes a "TSP" block and a "TSP" block. The model is titled "2 [0..3]".

23

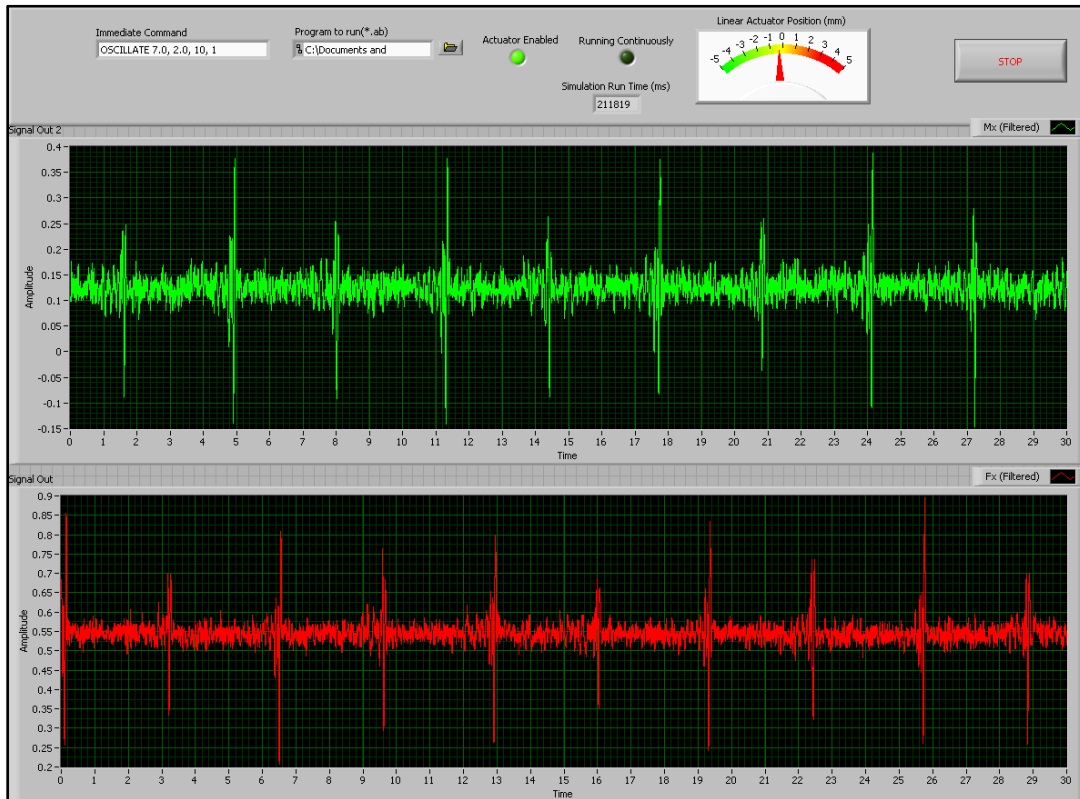


Figure 14: Screenshot of the LabVIEW control panel for the experimental setup

The force data is saved at each run of the actuator and this data is later imported into a MATLAB code to plot final results.

3.8 Testing Cases and Parameters

The experimental testing is conducted in three phases. The first phase characterizes baseline sloshing values for the fluid within the tank, i.e. free sloshing. The second phase repeats the experiment to evaluate the degree of damping present due to inactive micro-baffles floating on the free surface. The floating micro-baffles will be activated during peak sloshing in the third phase of testing. Each of the three phases will be repeated for two test cases; ‘low excitation’ sloshing and ‘high excitation’ sloshing. Low excitation sloshing corresponds to a condition such that the slosh waves reach a height that is 12.5% of the tank diameter. The actuator stage displacement is set at 1.8mm. Similarly, high excitation sloshing condition occurs when the slosh waves reach a height that is 25.0% of the tank diameter. The stage displacement is set at 3.0 mm, the frequency of the actuator is set at 2Hz for all the conditions, running for 12 cycles. This results in a condition where the actuator runs for six seconds and halts, allowing a remnant slosh condition beyond this point. Table I below outlines the summary parameters used for each test case.

Table 2: Summary of the parameters for the test condition

Test Conditions	Actuator Stage Displacement, a (mm)	Frequency, f_n (Hz)	Cycles, N_c
Low Slosh	1.8	2.0	12
High Sloshing	3.0	2.0	12

Using these test conditions, three cases for the experiment are conducted. The first case is a free slosh case, where the tank is agitated to provide sloshing due to the free surface of the liquid. The magnetic source is mounted to be immersed in the free surface in this case. This case is to provide a benchmark for the comparing the effect of using the magnetoactive micro-baffles. The second case is to place the micro-baffles on the free surface around the magnet setup and left to move freely. The micro-baffles cover close to 85% of the free surface. The magnetic source is at its OFF state at this point. This case is to visualize the damping effect of the simply placing the micro-baffles in the tank. Finally the third case is to activate the micro-baffles by keeping switching the micro-baffle to its ON state at the time of actuation. This will provide the final comparative results for the effect of magnetoactive micro-baffles.

Table 3 below shows a summary of the three cases that are considered for each test condition.

Table 3: Summary of the three testing cases

TESTING CASES	
CASE A	Free Slosh
CASE B	Slosh with inactive micro-baffles
CASE C	Slosh with active micro-baffles

Testing for both phases is started by generating the sloshing laterally until a natural frequency is reached, i.e. the peak of its sloshing activity. The agitation of the tank is then stopped to allow natural damping to occur. This is allowed for a period of time in order to achieve complete damping.

4. COMPUTATIONAL APPROACH

4.1 CFD Theory

4.1.1 An Introduction to CFD

The computational approach implements the field of Computational Fluid Dynamics (CFD). This is used here to compliment experimental results. This method comprises of the solution of Navier-Stokes equations at required points to get the properties of the fluid flow at those points. This technique exists since advancement in complex mathematical algorithms in 1930 [16]. Simple CFD problems were solved analytically, but with increase in fluid flow complexity, mathematical complexity increases exponentially. With the advancement of computers since 1950s, with powerful graphics and 3D interactive capability, use of CFD has gone beyond research and into industry as a design tool. Experiments can give macro data at certain points in the flow field, but with CFD, flow field can be resolved to details like turbulence, viscous forces and velocity. All this makes CFD an essential and useful tool for complex flows like propellant slosh.

CFD is carried out by employing the solution of Navier-Stokes equations. The Navier-Stokes equations are a set of partial differential equations describing the processes of momentum and heat and mass transfer. These equations have no known general analytical solution, but can be solved numerically by discretization. The four essential Navier Stokes equations that are commonly used are the x-momentum, y-momentum, z-

momentum and continuity equations and are respectively shown below in their conservation Cartesian coordinate form:

The Momentum Equations

X - Momentum

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u V) = -\frac{\partial p}{\partial x} - \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) \quad (2)$$

Y - Momentum

$$\frac{\partial(\rho v)}{\partial t} + \nabla \cdot (\rho v V) = -\frac{\partial p}{\partial y} - \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) \quad (3)$$

Z - Momentum

$$\frac{\partial(\rho w)}{\partial t} + \nabla \cdot (\rho w V) = -\frac{\partial p}{\partial z} - \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \right) \quad (4)$$

The Continuity Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho V) = 0 \quad (5)$$

4.1.2 Sloshing and CFD

Before the advances in Computational Fluid Dynamics (CFD) and computing power, propellant slosh issue was dealt on case by case basis. Experimental setup would be

designed to validate mathematically predicted parameters such as frequency and damping for propellant slosh. This process was time consuming and economically draining. The parameters thus obtained were used to design passive propellant management devices (PMD) like baffles and diaphragms for propellant slosh damping. These passive damping devices provide excellent propellant slosh damping for a small range of frequency and small amplitude of sloshing, but they are not effective when propellant fill level changes and sloshing frequency is outside their design range. These devices are bulky, consume space, add significant weight and have small operation range.

The CFD analysis of the slosh damping system using micro-baffles is categorized as a Fluid-Structure Interaction (FSI) with Electromagnetism problem. FSI problems and Multiphysics problems in general are often too complex to solve analytically and so they should be analyzed by means of experimentation or numerical simulation. Research in the fields of computational fluid dynamics is still ongoing but the maturity of this field enables numerical simulation of FSI. Two main approaches exist for the simulation of FSI problems: a monolithic approach where the equations governing the flow and the displacement of the structure are solved simultaneously using a single solver. The other is the partitioned approach where the equations governing the flow and the displacement of the structure are solved separately with two distinct solvers.

The tank utilized in this simulation is designed with the same dimensions as the experimental tank so that an accurate comparison can be made to the experimental simulations. The tank is set to act as the outer structural boundary of the system. For

simplicity purposes, the tank is modeled as a thin walled cylinder. The FSI system, like all subsequent tank models, is defined as a transient analysis because it is assumed that the behavior of the fluid is ever-changing with respect to time. The total simulation time is also carefully chosen so that the solution to the fuel slosh model is allowed to build in time and produce acceptable results. The simulation is programmed with a user-defined function to “excite” the propellant tank model at a particular frequency for a predetermined amount of time. As the tank oscillates about a single axis, the computational solver is recording the desired outputs of the particular test. For this simulation, the tanks are excited for six seconds before stopping, allowing the fluid to dampen for an additional period of time. Three simulations will be executed to match the three phases previously outlined in the experimental approach.

4.2 CD Adapco STAR-CCM+

STAR-CCM+ provides the world's most comprehensive engineering physics simulation inside a single integrated package. Much more than just a CFD solver, STAR-CCM+ is an entire engineering process for solving problems involving flow (of fluids or solids), heat transfer, and stress [17]. It provides a suite of integrated components that combine to produce a powerful package that can address a wide variety of modeling needs. It incorporates a 3D-CAD feature-based parametric solid modeler that allows geometry to be built from scratch. A major feature of 3D-CAD is design parameters, which allows the user to modify the 3D-CAD model from outside of 3D-CAD.

Simulations can be set up, run and post-processed from within popular CAD and PLM environments such as SolidWorks, CATIA V5, Pro/ENGINEER, and NX. STAR-CCM+'s unique approach an accurate CFD solution from a CAD model more quickly or more reliably. CFD results are linked directly to the CAD geometry (a process called associativity). After any modification in the CAD model the simulation results can be updated almost instantly by clicking the “update solution” button, allowing the rapid and thorough investigation of the design space.

At the heart of STAR-CCM+ is an automated process that links a powerful surface wrapper to CD-adapco's unique meshing technology. The surface wrapper significantly reduces the number of hours spent on surface cleanup and, for problems that involve large assemblies of complex geometry parts, reduces the entire meshing process to hours instead of days. STAR-CCM+'s single integrated process provides the fastest, most automatic route from complex CAD to CFD mesh. Advanced automatic meshing technology generates either polyhedral or predominantly hexahedral control volumes at the touch of a button, offering a combination of speed, control, and accuracy. For problems involving multiple frames of reference, fluid-structure interaction and conjugate heat transfer, STAR-CCM+ can automatically create conformal meshes across multiple physical domains.

In addition to its provision for inviscid and laminar flow, STAR-CCM+ has a comprehensive range of turbulence models such as k-epsilon (Standard, V2F, Realizable, Two-layer), k-omega (Standard, SST and BSL), Reynolds Stress (RSM – linear and

quadratic), Spalart-Allmaras Turbulence models, Boundary-layer transition, Large Eddy Simulation (LES), and Detached Eddy Simulation (DES, in the new Delayed Detached Eddy Simulation or DDES formulation).

Several third-party analysis packages can be coupled with STAR-CCM+ to further extend the range of possible simulations. Co-simulation is possible using Abaqus, GT-Power, WAVE and OLGA, and file-based coupling is possible for other tools such as Radtherm, NASTRAN and ANSYS.

This software is based on object-oriented programming technology. It is designed to handle large models quickly and efficiently using a unique client-server architecture that seamlessly meshes and simultaneously solves and post-processes over multiple computing resources without requiring additional effort from the user.

4.3 Process of Computational Analysis

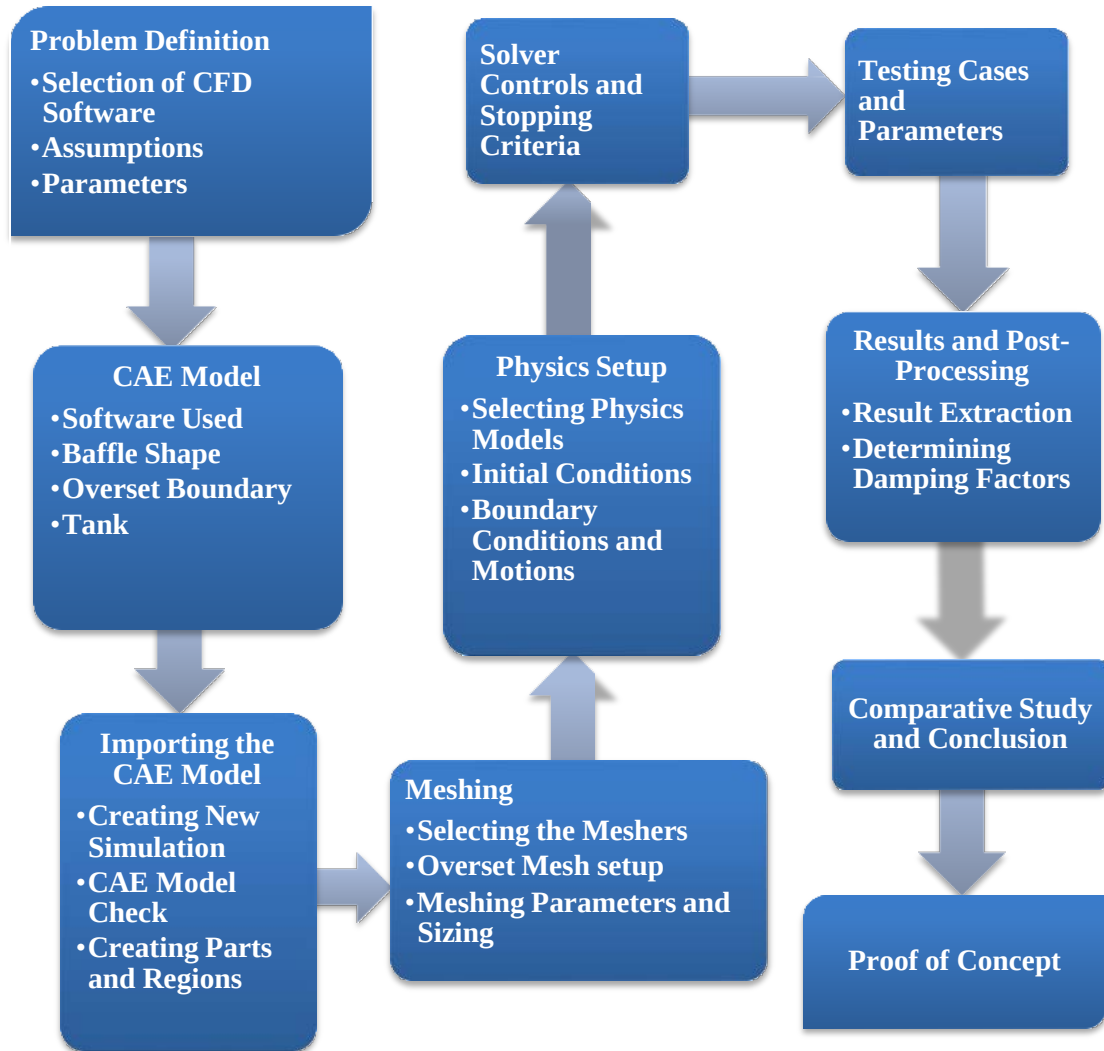


Figure 15: Process of computational analysis

4.4 Assumptions in Computational Approach

When comparing the conditions in a computational or theoretical domain with real life conditions there are many factors that are not taken into consideration. Hence there are assumptions in the computational approach that need to be taken in order to compare with the experimental approach. With new advances in CFD solvers most of these assumptions are not required or are reduced. The assumptions taken for this computational approach are:

- Due to the complexity of the CAE design, an array of micro-baffles is placed in the tank in a proper order as opposed to a random order of micro-baffles.
- The above assumption also shows a difference in the number of micro-baffles actually placed in the experimental tank compared to the number of micro-baffles constructed in the CAE model.
- The magnetic field from the magnetic activation source constrains the motion of the micro-baffles along the free surface of the water in the tank while the tank is in motion. This is used to assume that the micro-baffles in the computational model are rigid bodies that are constrained from its six degrees of freedom.
- The thickness of the wall of the tank is not considered.
- The hollow channels are not considered in the micro-baffle design as they are interior sections and are not required for meshing.
- The total weight of the micro-baffle (base, lid and steel nails) is considered and used for rigid body calculation.

- A Volume of Fluid Method is adapted to define the free surface by using an isosurface.

4.5 CAD Design

The CAD design of the micro-baffles and the tank are made using CATIA V5. The dimensions of each micro-baffle are set to match the dimensions of the physical model. Based on the assumptions of the computational approach the micro-baffles are aligned to make a semi-hexagonal grid as seen in Figure 16. The micro-baffles are spaced from each other for dynamic meshing quality.

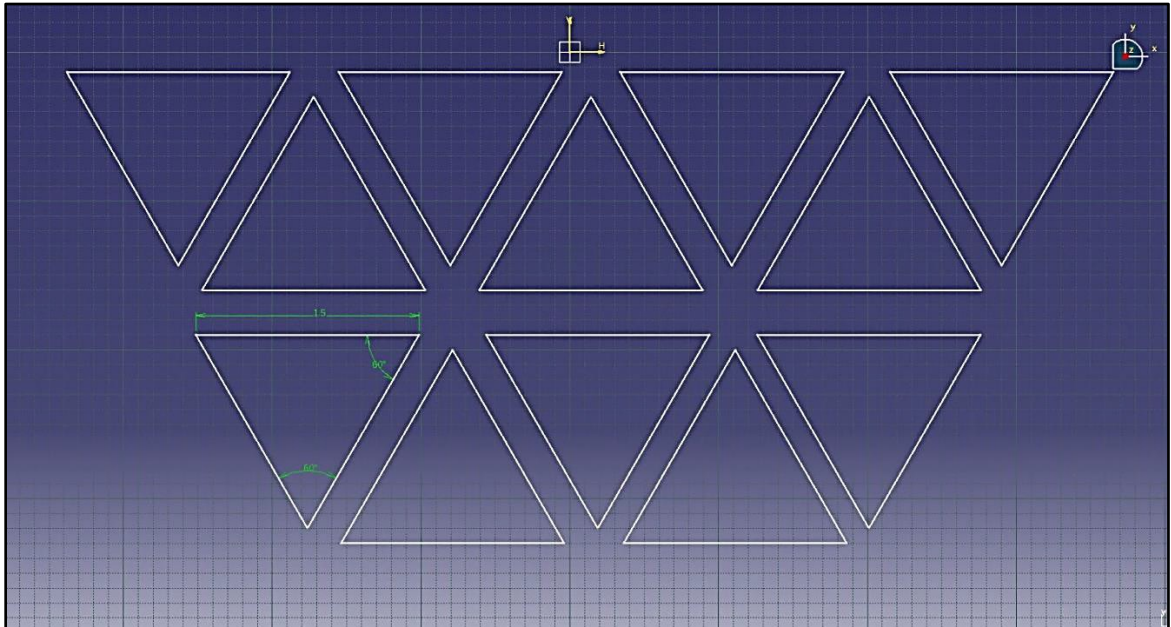


Figure 16: CATIA V5 model for the micro-baffles

Figure 17 shows the CAE Model for the tank. The CAE model is a simple semi-cylinder of the same dimensions as the experimental tank. This will act as the main domain for the

simulation. Only half of the cylinder is considered as a symmetrical behavior is assumed. This results in a vast reduction in the simulation time. The combination of these designs is imported into Star CCM+ using its 3D-CAD import feature.

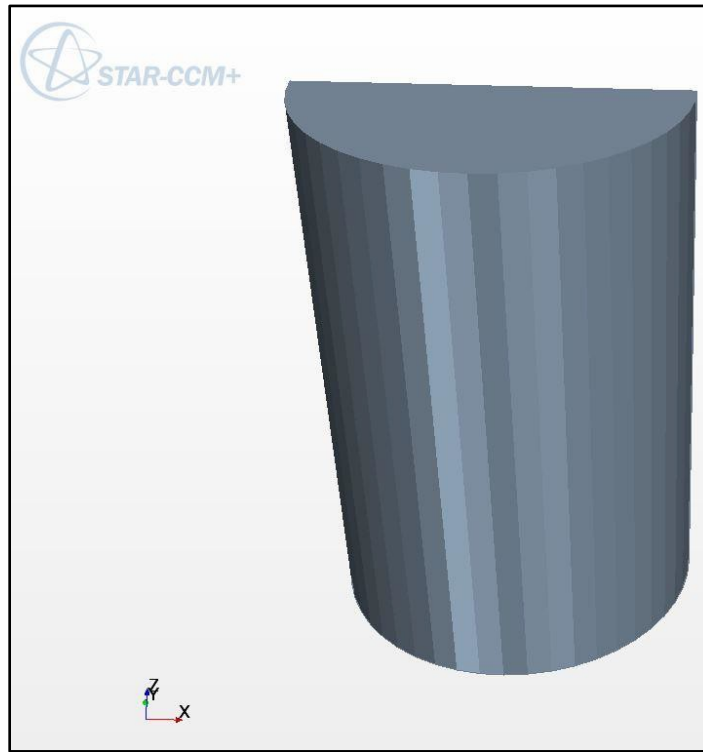


Figure 17: CATIA V5 model for the slosh tank

4.6 Meshing process

A mesh is the discretized representation of the computational domain, which the physics solvers use to provide a numerical solution. STAR-CCM+ provides meshers and tools that are used to generate a quality mesh for various geometries and applications. The following meshing strategies are available:

Unstructured Meshing

- a. Parts-Based Meshing - Parts-based meshing detaches the meshing from the physics and provides a flexible and repeatable meshing pipeline. This strategy provides several advantages over region-based meshing.
- b. Region-Based Meshing - Region-based meshing meshes the surface and volume at the region-level. Only use region-based meshing if you intend to use a mesher that is not available in parts-based meshing.

Structured Meshing - Directed Meshing

Using any of these strategies the following meshers are commonly in STAR-CCM+:

Surface Meshing

Surface Wrapper	Wraps the initial surface to provide a closed and manifold surface mesh from complex geometry.
Surface Remesher	Remeshes the initial surface to provide a quality discretized mesh that is suitable for CFD.

Volume Meshing

Trimmed Mesher	Generates a volume mesh by cutting a hexahedral template mesh with the geometry surface.
Polyhedral Mesher	Generates a volume mesh that is composed of polyhedral-shaped cells.

Tetrahedral Mesher	Generates a volume mesh that is composed of tetrahedral-shaped cells.
Prism Layer Mesher	Adds prismatic cell layers next to wall boundaries. The mesher projects the core mesh back to the wall boundaries to create prismatic cells.
Advancing Layer Mesher	Creates a volume mesh composed of prismatic cell layers next to wall boundaries and a polyhedral mesh elsewhere. The mesher creates a surface mesh on the wall and projects it to create the prismatic cell layers.
Extruder Mesher	Generates an extruded mesh region from a boundary that one of the core volume meshers has meshed.
Thin Mesher	Generates a prismatic layered volume mesh for thin geometries.
Generalized Cylinder Mesher	Generates a volume mesh appropriate for elongated cylindrical regions.

Structured Meshing

Directed Meshing	Generates a mesh by creating a two-dimensional patch mesh on a surface and sweeping it through the volume along a specified guide surface.
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The Mesh model for the micro-baffles in the tank incorporates a Surface Remesher and Trimmed Mesher by parts-based meshing, with an applied volumetric control on the free surface of the micro-baffle. This is applied for each individual micro-baffle to provide a

fine resolution in the mesh. The region in tank near the water surface is also refined to provide a more accurate solution of the Volume-Of-Fluid (VOF) as shown in Figure 18.

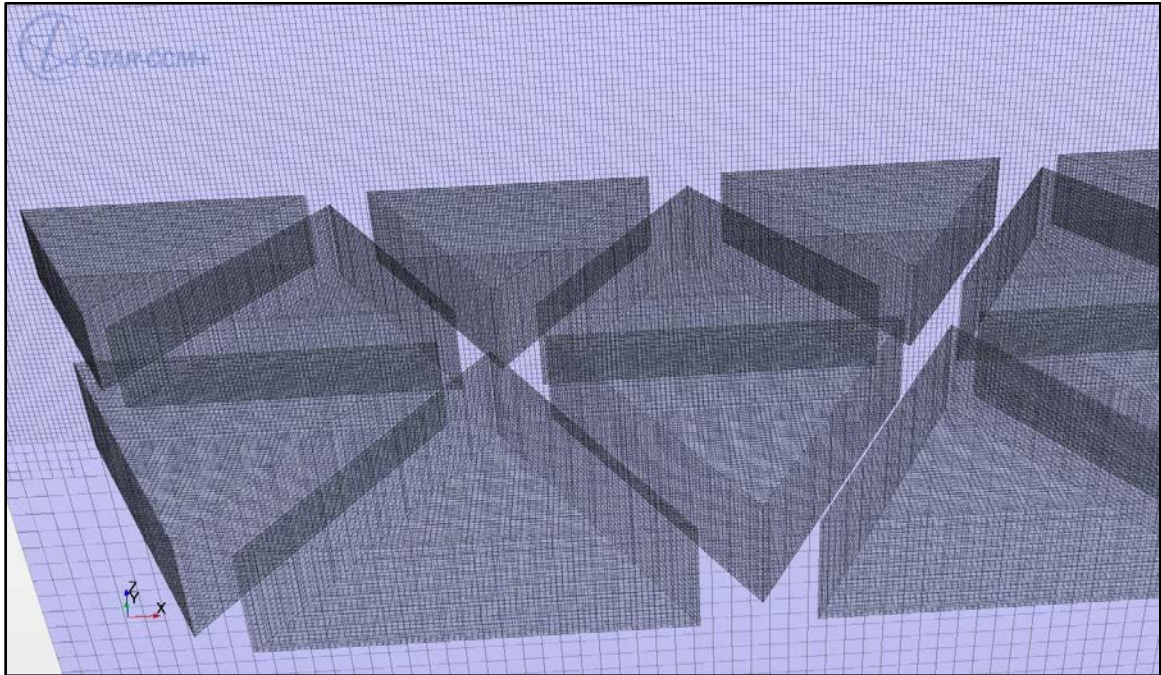


Figure 18: Close view of the mesh in the near micro-baffle region

Parts-based meshing is a meshing strategy which creates one or more mesh operations to define the steps that STAR-CCM+ performs to generate the volume mesh. The parts meshes continua represents the volume mesh that you generated for the part as a continuum that the physics region can use. Conceptually, the parts meshes continua is a bridge between the parts-based mesh and the physics regions. This is the method implemented.

4.7 Setting up the Physics

Upon generating a suitable volume mesh, a physics model is setup. In STAR-CCM+ this is also referred to as setting up the physics continuum. Generally, a physics continuum is defined as continuous and nonspatial, such that no part of it is distinct or distinguishable from adjacent parts. It logically interprets a continuum as a collection of models that represents the substance (fluid or solid) being simulated. STAR-CCM+ offers a wide choice of models for simulating flow and energy when the continuum mechanics formulation of fluid dynamics is valid.

The physics models used in this computational analysis are shown below:

1) Space Model

The Three-Dimensional model is designed to work on three-dimensional meshes and for the current sloshing problem. This model is activated only if the mesh is indeed three-dimensional. The use of a one-cell-thick three-dimensional mesh is much less efficient than using a true two-dimensional mesh for two-dimensional and axisymmetric simulations. A better approach is to extract a two-dimensional mesh.

2) Eulerian Multiphase Model

Multiphase flow is a term which refers to the flow and interaction of several phases within the same system where distinct interfaces exist between the phases. The term ‘phase’ in broader terms, can be defined as a quantity of matter within a system that has

its own physical properties to distinguish it from other phases within the system. For example it relates to liquids of different density, bubbles of different size, or Particles of different shape.

Volume of Fluid (VOF), a simple multiphase model, is used for this simulation. It is suited to simulating flows of several non- immiscible fluids on numerical grids capable of resolving the interface between the phases of the mixture, as shown below. In such cases, there is no need for extra modeling of inter-phase interaction, and the model assumption that all phases share velocity, pressure, and temperature fields becomes a discretization error. Due to its numerical efficiency, the model is suited for simulations of flows where each phase constitutes a large structure, with a relatively small total contact area between phases. A good example of this type of flow is sloshing flow in a water tank, where the free surface always remains smooth. If the tank movement becomes pronounced, this results in breaking waves, large numbers of air bubbles in the water, and water droplets in the air. The method would then require a fine mesh (at least three cells across each droplet/bubble) to produce small modeling errors. The spatial distribution of each phase at a given time is defined in terms of a variable that is called the volume fraction. A method of calculating such distributions is to solve a transport equation for the phase volume fraction The method uses the STAR-CCM+ Segregated Flow Model.

3) Gravity Model

The gravity model accounts for the action of gravitational acceleration in STAR-CCM+ simulations. For fluids, it provides two effects:

- The working pressure becomes the piezometric pressure.
- The body force due to gravity can be included in the momentum equations.

4) *Time Model*

The Implicit Unsteady model is the only unsteady model available with the Segregated Flow and Segregated Fluid Energy models. It uses the Implicit Unsteady solver. When this model is activated, objects that offer a choice between Iteration or a time-step for a trigger can be set to update at each time-step.

5) *Flow Model*

The Segregated Flow model solves the flow equations (one for each component of velocity, and one for pressure) in a segregated, or uncoupled, manner. The linkage between the momentum and continuity equations is achieved with a predictor-corrector approach. This model has its roots in constant-density flows. Although it can handle mildly compressible flows and low Rayleigh number natural convection, it is not suitable for shock-capturing, high Mach number, and high Rayleigh-number applications.

6) *Turbulence Models*

In this simulation, a K-Epsilon turbulence model is used. A K-Epsilon turbulence model is a two-equation model in which transport equations are solved for the turbulent kinetic energy and its dissipation rate. Various forms of the K-Epsilon model have been in use for several decades, and it has become the most widely used model for industrial

applications. Since the inception of the K-Epsilon model, there have been countless attempts to improve it. The most significant of these improvements have been incorporated into STAR-CCM+.

The turbulence models in STAR-CCM+ are responsible for providing closure of the governing equations in turbulent flows. This deals with models that provide closure for the Reynolds-Averaged Navier-Stokes (RANS) equations. To obtain the Reynolds-Averaged Navier-Stokes (RANS) equations, the Navier-Stokes equations for the instantaneous velocity and pressure fields are decomposed into a mean value and a fluctuating component. The averaging process may be thought of as time averaging for steady-state situations and ensemble averaging for repeatable transient situations. The resulting equations for the mean quantities are essentially identical to the original equations, except that an additional term now appears in the momentum transport equation.

7) *Motion Models*

In addition to the flow models, the motion models of the tank and micro-baffles are set. There are three broad categories for defining motion in STAR-CCM+:

- a. The first category contains those methods that involve actual displacement of mesh vertices in real time: use these methods with a transient analysis. Several methods are available in this category:
 - Rigid Motion

- Rotation
 - Translation
 - Rotation and Translation
 - Morphing (deforming mesh)
 - Dynamic Fluid Body Interaction (DFBI)
 - DFBI Rotation and Translation
 - DFBI Embedded Rotation
 - DFBI Morphing
 - DFBI Superposed Rotation
 - Solid Displacement
 - User-Defined Vertex Motion
- b. The second category contains the method whereby an unsteady motion problem is converted to a steady-state problem by imposing a moving frame of reference on a static mesh. Rigid rotation and/or translation can be simulated in this way. The reference frame types that can be used in this category are Rotating, and Rotating and Translating. The third type of motion is a corollary of the real-time displacement, and applies only to harmonic balance flutter. The imposed modal displacement of the flutter boundary is applied to the frequency domain solver. There is only one option for this approach.

For this simulation Category 1 motion methods are applied, specifically the ***Translation*** motion for the tank and the ***DFBI Rotation and Translation*** as shown in Figure 19.

Motions	
DFBI Rotation and Translation	
High Amplitude Actuation	
Translation Velocity	$u_{highamp}$
Coordinate System	Laboratory
Managed Coordinate Systems	
Superposing Motions	
Low Amplitude Actuation	
Translation Velocity	u_{lowamp}
Coordinate System	Laboratory
Managed Coordinate Systems	
Superposing Motions	
Stationary	

Figure 19: Motion model setup

The translation motion for the tank is applied on the walls of the tank. A field function is used to define this motion, and it corresponds to velocity values tabulated in a table. The table values are based on the Simple Harmonic Motion (SHM) values of velocity given the amplitude and frequency values. The equation for velocity, v , of a body in SHM is given below:

$$v = 2\pi af \cos(2\pi ft) \quad (6)$$

Where, f is the frequency of oscillation, and a is the amplitude of the oscillation.

The DFBI (Dynamic Fluid Body Interaction) module is used to simulate the motion of a rigid body (micro-baffle) in response to pressure and shear forces the fluid exerts, and to additional forces defined. The STAR-CCM+ solvers calculate the resultant force and moment acting on the body due to all influences, and solves the governing equations of

rigid body motion to find the new position of the rigid body. Multiple rigid bodies can be modeled, with optional body couplings between them. The DFBI Rotation and Translation model DFBI Rotation and Translation solves the equations of rigid body motion for all 6-DOF Bodies. The individual micro-baffle parts are assigned as shown in Figure 20.

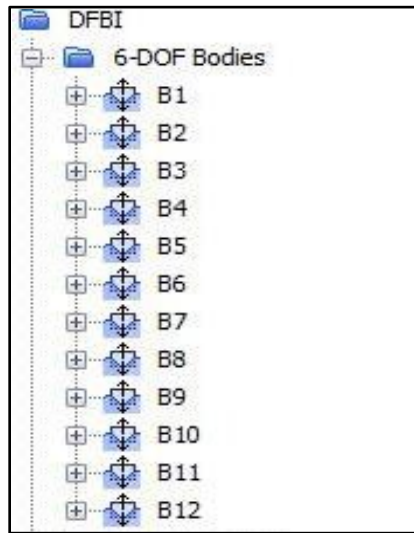


Figure 20: DFBI 6-DOF bodies assigned to the micro-baffle parts

The 6-DOF motion solver moves the vertices of the grid according to motion of the body that is calculated by the 6-DOF solver. The 6-DOF solver computes fluid forces and moments and gravitational forces on a 6-DOF body. Pressure and shear forces are integrated over the surfaces of the 6-DOF bodies. More forces and moments can be added. The forces and moments acting on the 6-DOF body are used to compute the translational motion of the center of mass of the body and the angular motion of the orientation of the body. The input coordinates of the individual rigid bodies and their

corresponding masses, and the mass moment of inertia of the triangular prism. Refer to Appendix A: Rigid Body Coordinates of the Baffles for DFBI Setup for the full rigid body coordinates.

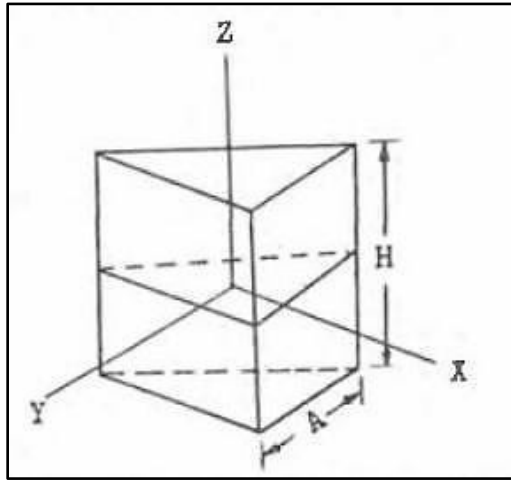


Figure 21: A triangular prism

For the micro-baffles, based on isosceles triangular prism geometry, the mass moment of inertia in the corresponding x, y and z directions are given as:

$$I_x = I_y = \frac{W}{24}(A^2 + 2H^2) \quad (7)$$

$$I_z = \frac{WA^2}{12} \quad (8)$$

Where, A is the side length, H is the height, and W is the weight of the prism.

A summary of the physics models used are shown below:

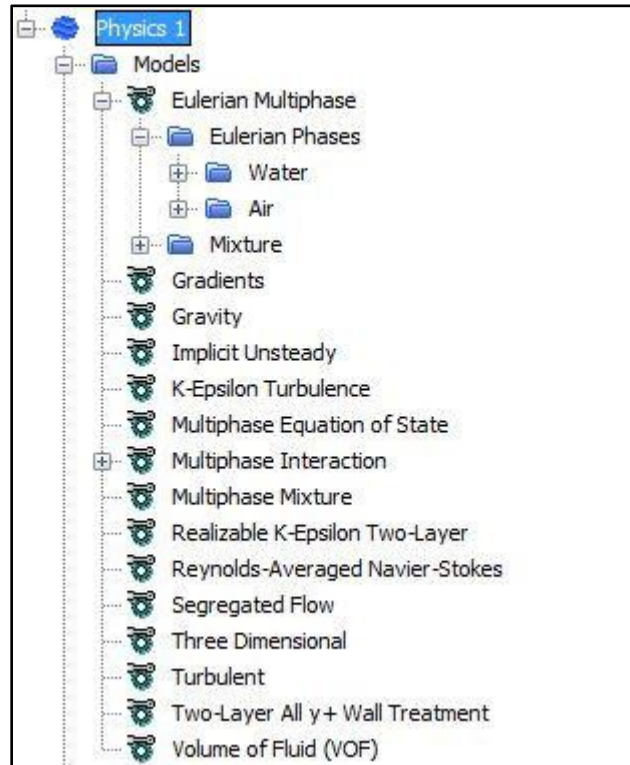


Figure 22: The physics models selected in the physics continua

4.8 Setting the Regions and Boundary Conditions

The mesh continuum models cannot be used for geometry parts. In order to mesh geometry parts, they must be assigned to one or more regions, and the mesh is then generated for the region. A part must be assigned to a region before its part surfaces can be assigned to boundaries, and its part curves to feature curves, within the region. The relationships do not need to be one-to-one; multiple parts can be assigned to the same boundary, and multiple part curves to the same feature curve.

Individual part surfaces cannot be assigned to a boundary until the parent part has been assigned to a region first. As for parts and regions, a bi-directional relationship exists between part surfaces and boundaries. A single region is created for the tank domain with each of the surfaces within this region specified. The top of the tank region represents an open region and is set as a pressure outlet boundary. The reference values of the pressure are maintained at the default assuming room temperature and pressure conditions. The individual baffle surfaces are labeled as shown in the figure below and are linked to the DFBI settings. However, within the tank, they are set as a wall with default settings. The wall and bottom surfaces of the tank are also set to the same conditions. A symmetry plane is set at the cut plane of the tank. This is to provide the symmetry conditions to project the second half of the cylinder.

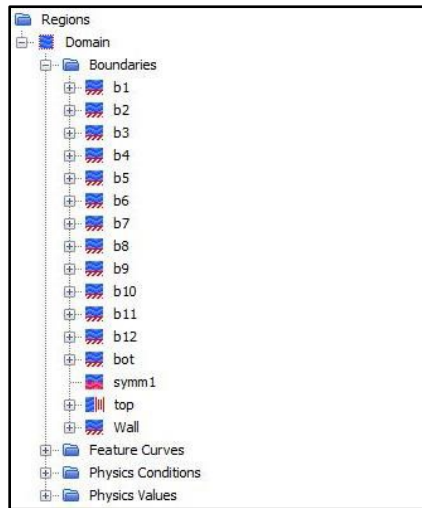


Figure 23: Screenshot showing the boundary setup

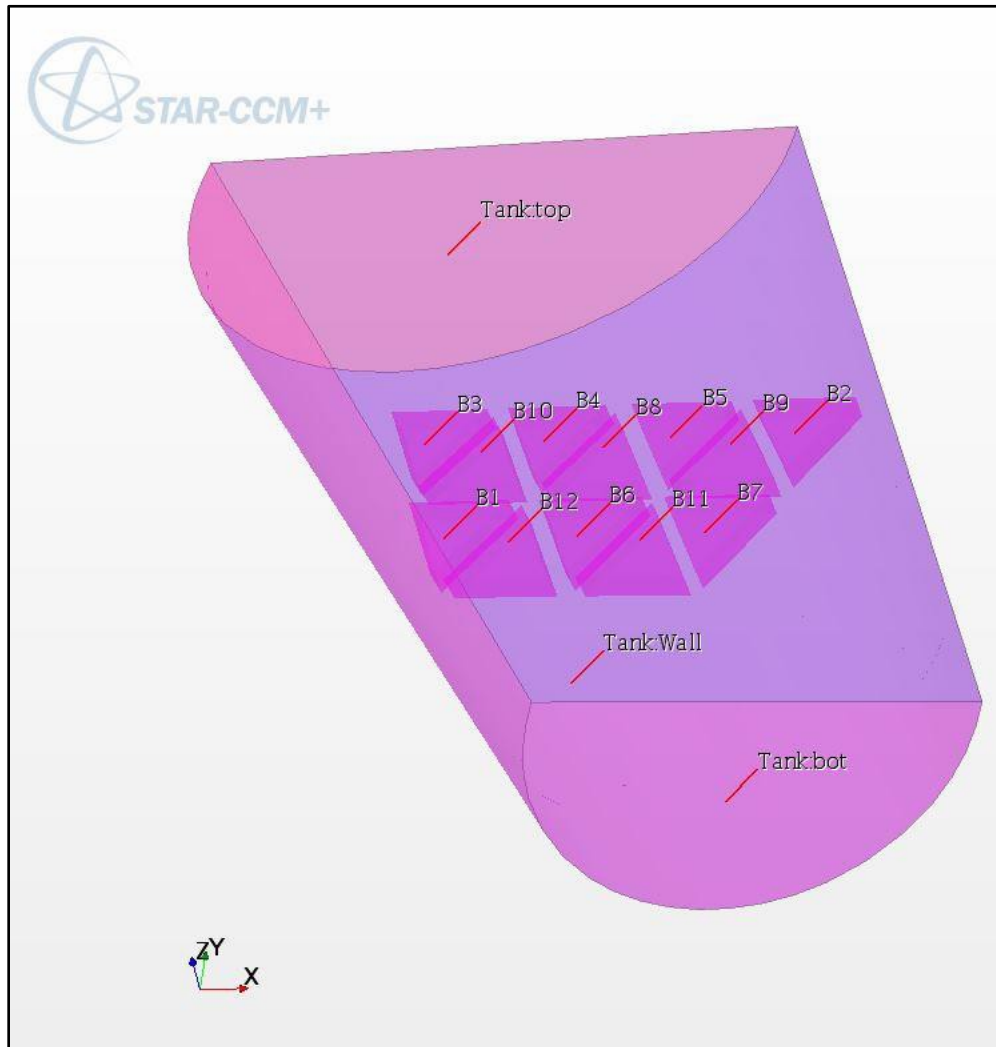


Figure 24: Screenshot of the domain and its regions

4.9 Setting the Initial Conditions

Initial conditions in the continua specify the initial field data for the simulation. For steady-state simulations, the converged solution should be independent of the initial field. However, the path to convergence, and hence the computational effort that is required to

reach convergence, is affected. Therefore, choose the initial conditions and values judiciously, particularly when the physics is complex.

Each model requires sufficient information for the primary solution data for the primary variables that are associated with the model to be set. In most cases, the initial conditions are set up directly. For some models, such as turbulence, there is the option of specifying the information in a more convenient form (turbulence intensity and turbulent viscosity ratio instead of turbulent kinetic energy and turbulent dissipation rate).

The following settings are applied in the continua

- The VOF height of water in the tank is set in the initial conditions for the tank region using a field function (Appendix B)
- Initial velocity for the tank wall motion and the rigid bodies are set to zero.
- Initial pressure values are constant and are set to the reference pressure values, based on the standard pressure conditions.
- The turbulence initial conditions are set at their default values.

4.10 Solver Setup and Running the Simulation

Most solvers that are seen within the Solvers node of a simulation tree are closely related to the models selected in a continuum, and are documented in the relevant modeling section. Transient analysis uses the unsteady SIMPLE algorithm. The regular segregated flow solver is used to perform inner iterations within each time step. The transient

phenomenon being simulated generally governs the physical time-step. For periodic flow, an order of 50–100 time steps per period is appropriate. In this case the time step is set at 0.001s.

The convective Courant number is used as a helpful indication for selecting the time step size: for time-accurate simulations, the convective Courant number should be 1.0 on average in the zone of the interest. This value implies that the fluid moves by about one cell per time step. For flows with free surfaces, if a second-order scheme for time integration is used, the Courant number must be less than 0.5 in all cells. If a free surface moves more than half a cell per time step, the High-Resolution Interface-Capturing scheme can lead to overshoots or undershoots, and eventually to divergence.

Stopping criteria allow you to specify how long the solution runs for and under what conditions it stops iterating and/or marching in time. Each specified stopping criterion is evaluated at the completion of every simulation step and a logical rule is used to determine if the interaction of all of the criteria stops the solver.

For unsteady simulations, there are four criteria:

- ***Maximum Inner Iterations*** - Stopping criterion is based on the number of inner iterations that the solver executes for transient analyses. The node of this criterion has its own properties, and appears when the implicit unsteady model is chosen. The values are set at 5.

- **Maximum Physical Time** - Stopping criterion is based on the simulation time that has elapsed in a transient analysis. The node of this criterion has its own properties, and appears when either the implicit unsteady model or the explicit unsteady model is chosen. This is set to 10 seconds for the simulation.
- **Maximum Steps** - Stopping criterion allows you to specify the maximum number of iterations in a steady solver or the maximum number of time-steps in an unsteady solver. Default Values are used for this simulation.
- **Stop File** - Criterion allows you to specify the pathname of a file (named ABORT by default) that, once in place, causes the solver to stop. This criterion is not used for this simulation.

Upon setting the solver and the stopping criteria, the solution is initialized and set to run. The residuals are observed using the monitors, and the scene is observed for the visualization of the simulation.

4.11 Post Processing Setup

STAR-CCM+ provides methods to analyze the solution while the simulation is running, as well as when it completes. Field functions provide access to raw simulation data, and to information computed from the raw data. These field functions can be analyzed using three general strategies:

- Creating reports of specific quantities. Four different types of report can be created, including one that accepts a user-defined expression in the same format

as a user-defined field function. In order to obtain a history of the reported value, reports can be monitored while a simulation is running.

- Visualizing the solution data in scalar or vector scenes. If these scenes are created before the simulation is run, they are updated continuously as the solution progresses.
- Plotting sets of data that are extracted from the solution. Monitors and residual plots, XY plots, and histogram plots may be created.

Figure 25, below shows the scene set to visualize the solution and to monitor the run of the simulation. Within the scene, a velocity and force monitor plot is added, along with an annotation of the time step and solution time.

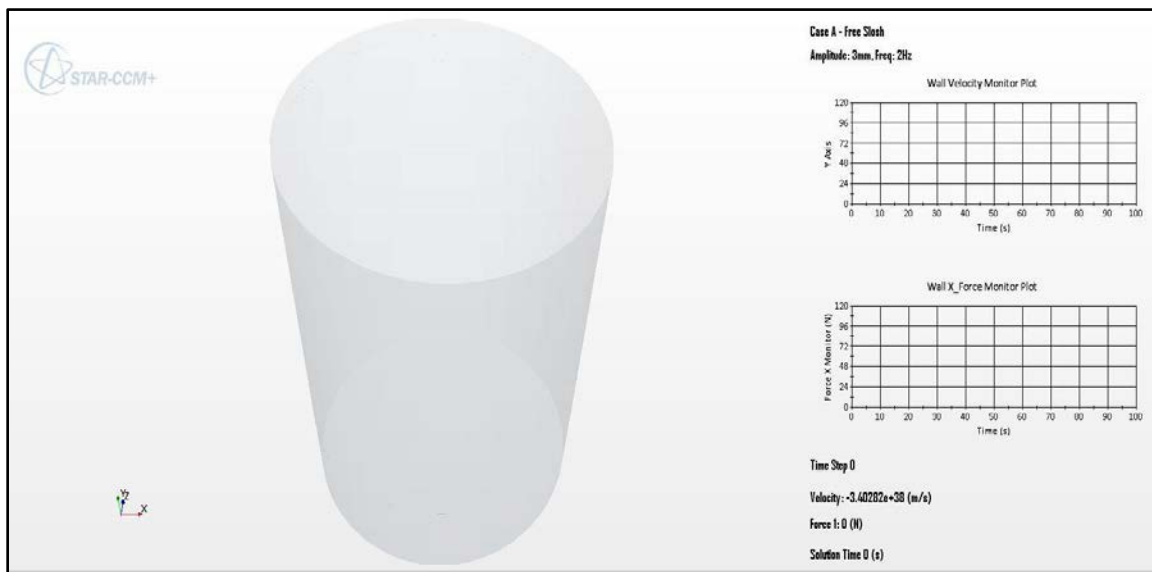


Figure 25: Scalar scene showing the tank geometry, monitors and solution time

For each of these strategies, parts from which the solution data is extracted are created. These parts can be regions, boundary surfaces, or derived parts, depending on the type of analysis. Derived parts are additional lines and surfaces that you create in addition to the surfaces already present on geometry parts and regions. A derived surface threshold is created on the surface of the tank wall close to the water surface level, similar to the height at which the load cells exist in the experimental setup.

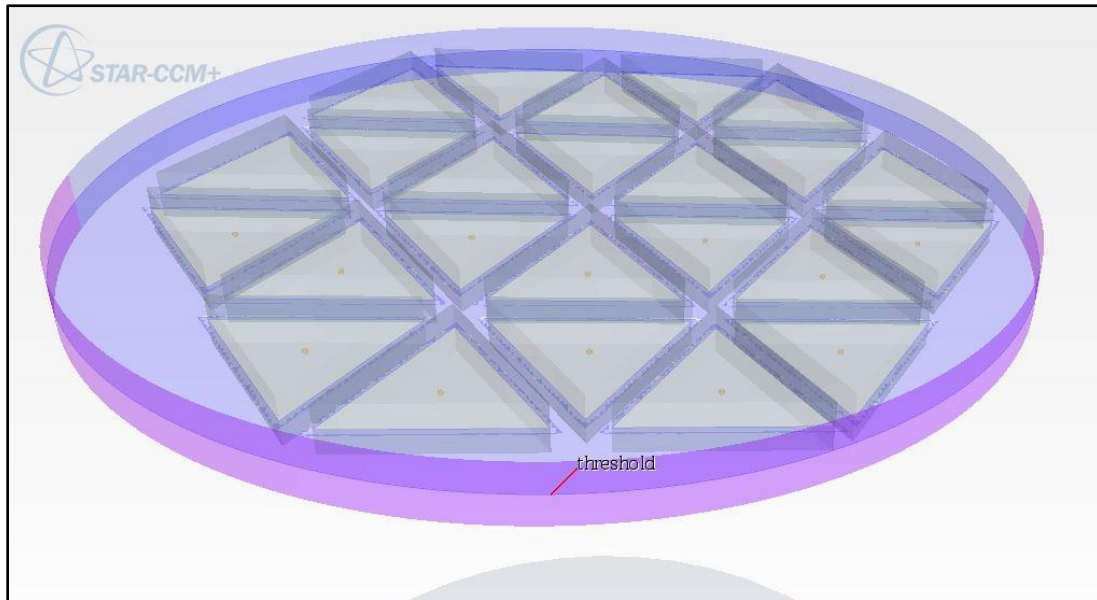


Figure 26: The derived part threshold used for force detection

Monitors are used to sample and save solution data while the simulation is running. Some monitor types are made available in response to the solvers are made active. Additional monitors can be created from a report.

Several features of STAR-CCM+ require that a trigger is defined for when an update event takes place.

5. RESULTS

Upon execution of the different cases of testing in the experimental approach, the load cell data is obtained and exported to a format in order to plot using MATLAB. Similarly upon completion of the CFD simulation the force data is exported from the post processed force plot and imported further plotted in the same manner.

The plots are further analyzed to determine the damping factor (ε) at the remnant slosh period. The amount of damping present is characterized by the time it takes the remnant sloshing to dissipate and return to equilibrium and this is defined by the damping factor:

$$\varepsilon = \frac{\delta}{\sqrt{(2\pi + \delta^2)}} \quad (9)$$

Where, δ is the logarithmic decrement (Appendix C)

Case A: The Free Slosh

For the free slosh case, the results of the CFD model show a strong correlation to the experimental results, which in turn also correlates well with the experimental results from past research efforts. It is interesting to note that the results from the CFD model create a cleaner looking than the experimental data. This is due to the fact that the experimental setup is not in a perfect environment. Since the CFD model is purely computational, the results are not influenced by external noise such as vibrations in the support cables and small amplitude excitation in the off axis direction.

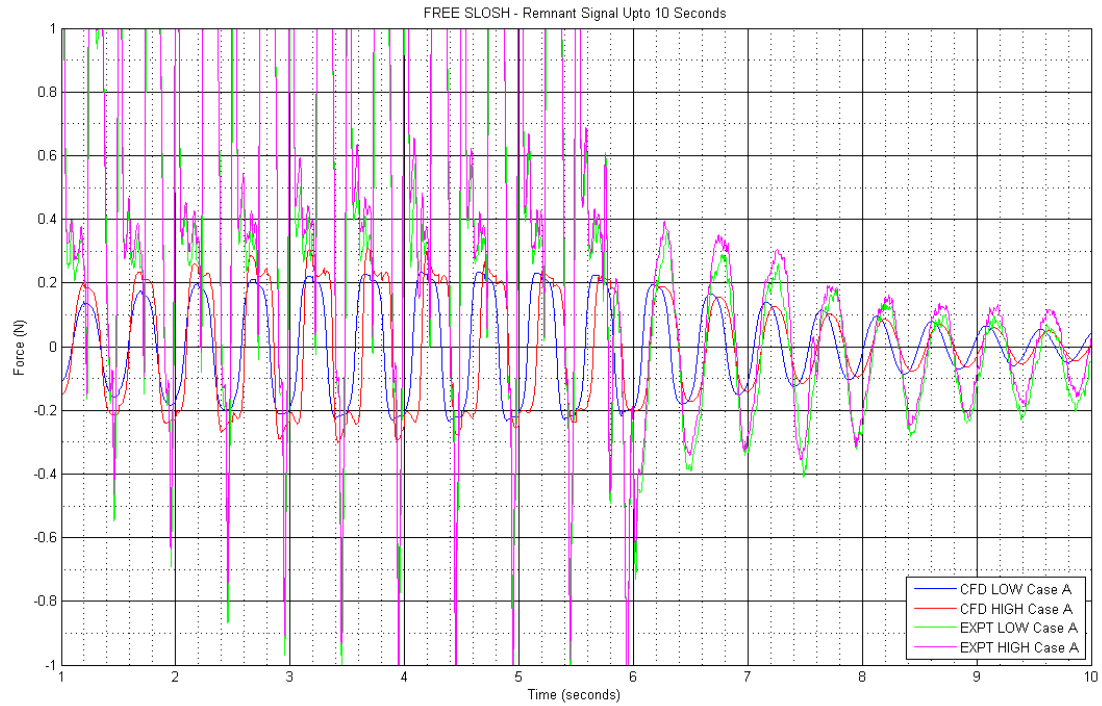


Figure 27: Results for the experimental and CFD at the all conditions of free slosh

Figure 28 shows a comparative plot between the experimental and CFD results for the free slosh case under the low amplitude condition. This refers to the 1.8mm setting for the excitation actuator stage displacement. The damping values for the experimental and CFD method are found to be 0.154 and 0.277 respectively. By observation, the peak occurrences (or the natural frequency) appear to be correlating well.

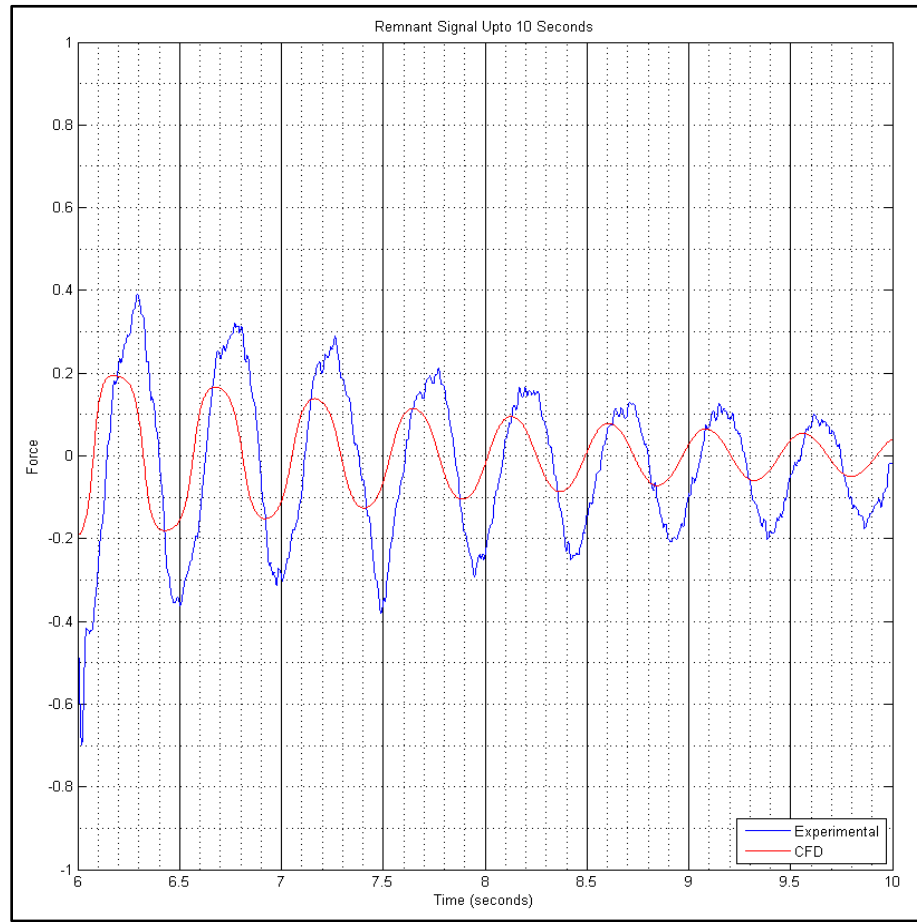


Figure 28: Comparative plot for the free slosh case at low excitation

During high excitation slosh conditions, the behavior of free slosh changes. In a realistic and theoretical environment, at this peak excitation frequency the remnant slosh conditions give larger initial remnant peaks that dampen at a higher ratio. Figure 29 shows the remnant slosh conditions for the experimental and CFD

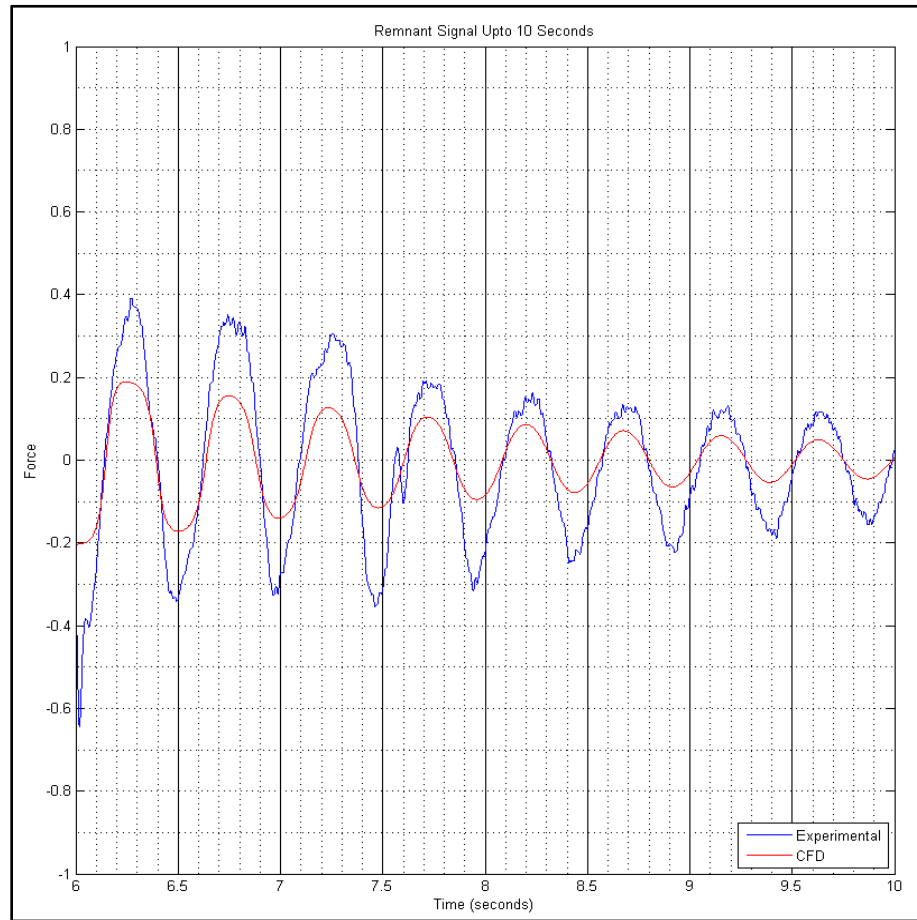


Figure 29: Comparative plot for the free slosh case at high excitation

From these results a damping factor of 0.221 and 0.298 is obtained for the experimental and CFD approach respectively.

Case B: The Addition of Micro-Baffles without Electromagnetic Actuation

In this case, the baffles are added to the setup of the tank with the electromagnet already mounted. However, the electromagnet is kept at its OFF state. This is termed as the inactive micro-baffle state. The CFD simulation for this case is not carried out due to high requirements in simulation time. Such a simulation also requires a complete 6 Degrees of Freedom (DOF) solution for each individual baffle with a high resolution mesh. The results for case 2 are compared to the experimental free slosh results.

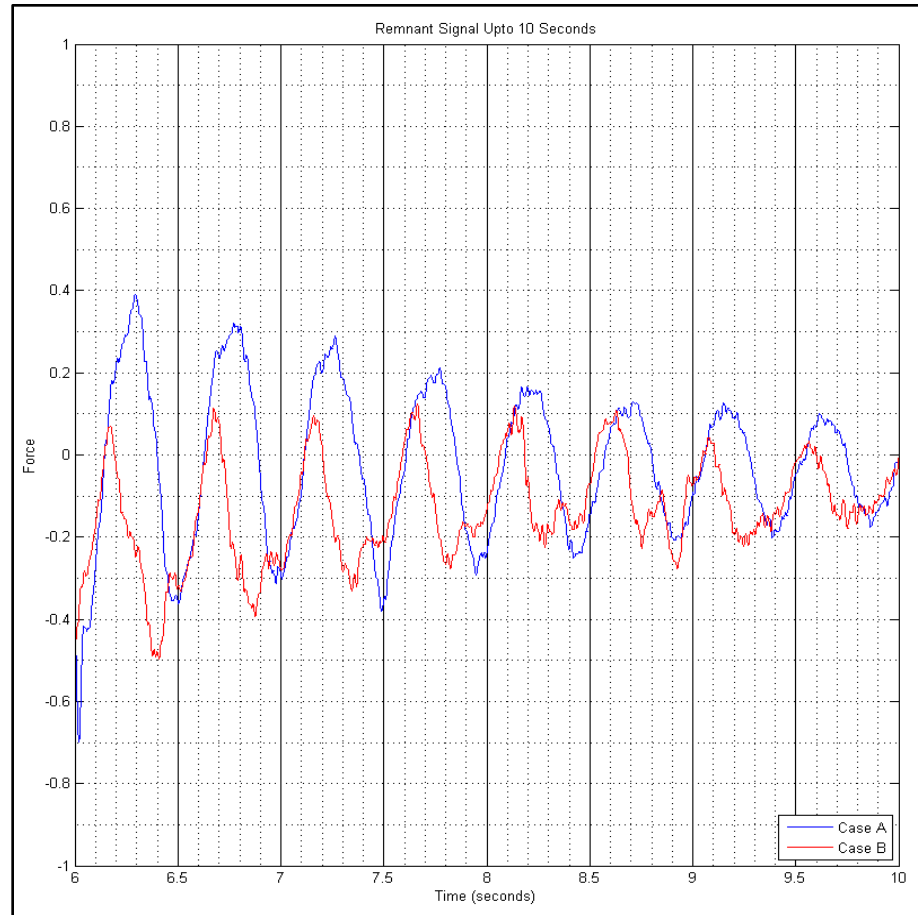


Figure 30: Comparison of Case A and Case B at low excitation slosh

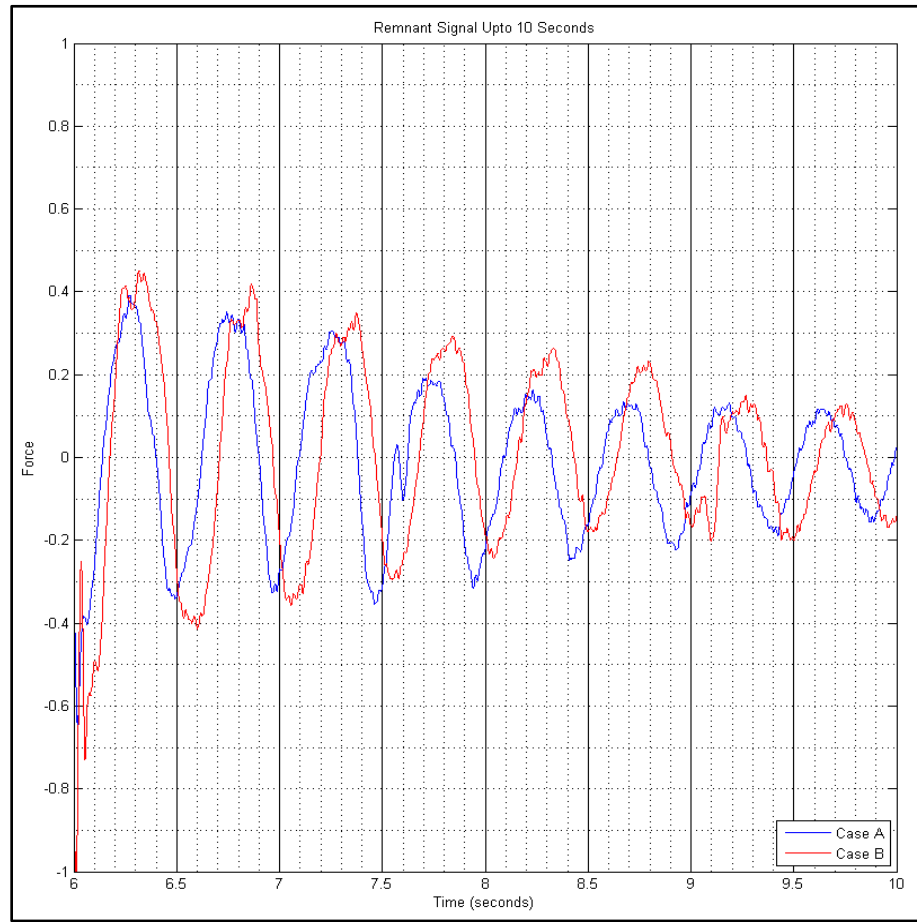


Figure 31: Comparison of Case A and Case B at high excitation slosh

The addition of the micro-baffles show a general increase in the damping factor from 0.154 to 0.195 in the low excitation slosh case and 0.227 to 0.258 for the high excitation slosh case. This evidently shows an improvement in the model upon the simple addition of the micro-baffles. However, in the experimental result, the remnant slosh shows an offset of the entire signal from the x-axis. This is due to inertial effects of the micro-baffles that are observed over many trials. As the peaks get smaller, the values converge to the x-axis and continue along the x-axis.

Case C: Magnetoactive Micro-Baffle Testing

Upon activating the micro-baffles by turning the electromagnet to its ON state the damping factors for each case are prominently higher and are observed in the plots.

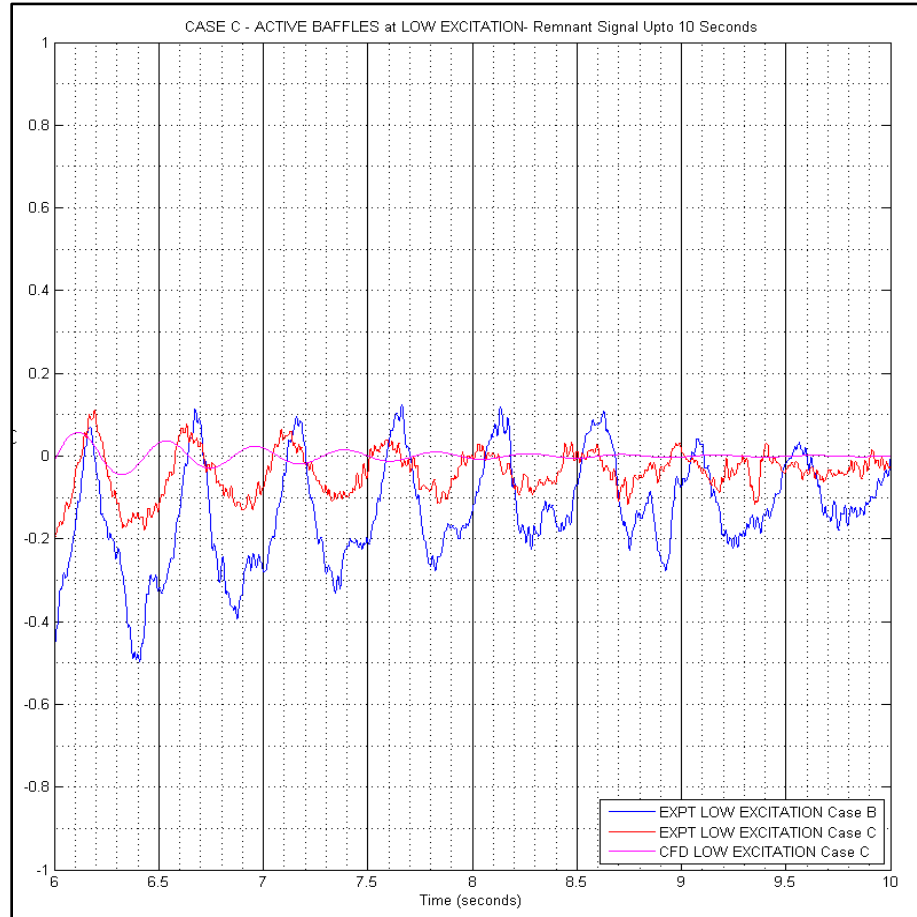


Figure 32 and Figure 33 show the comparison between the experimental results of the inactive and active micro-baffles, and the CFD case of the active micro-baffles.

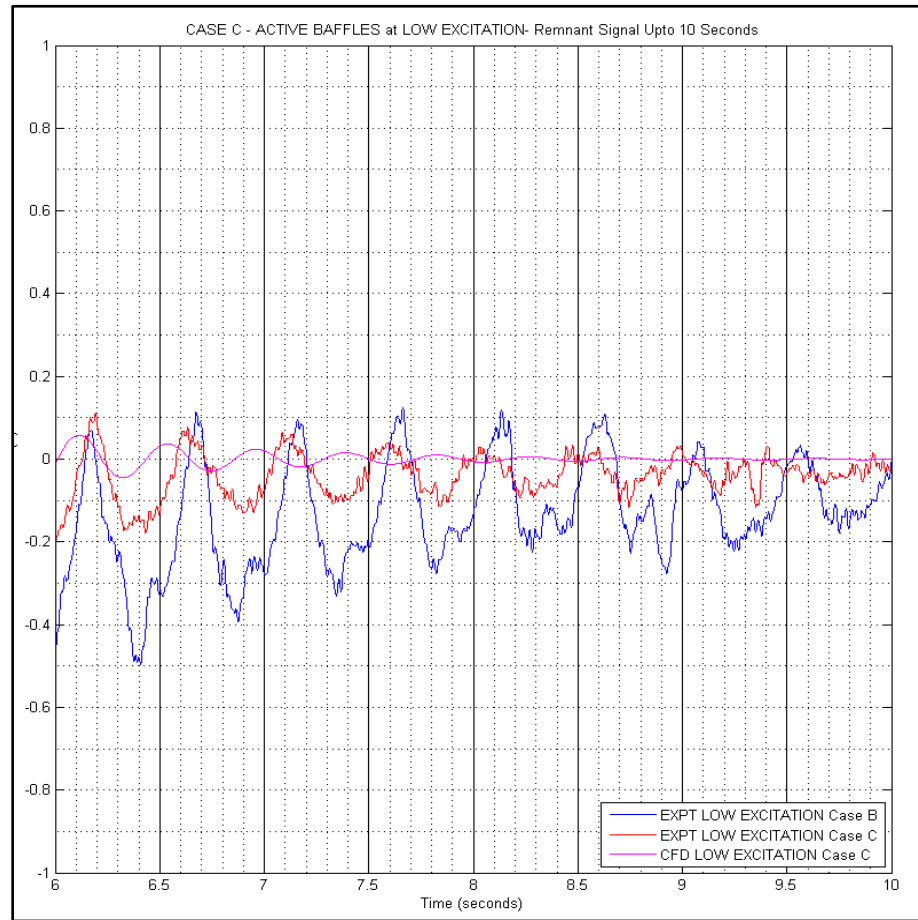


Figure 32: Comparative plot of experimental results from Case B and C, and the CFD results from Case C at low excitation

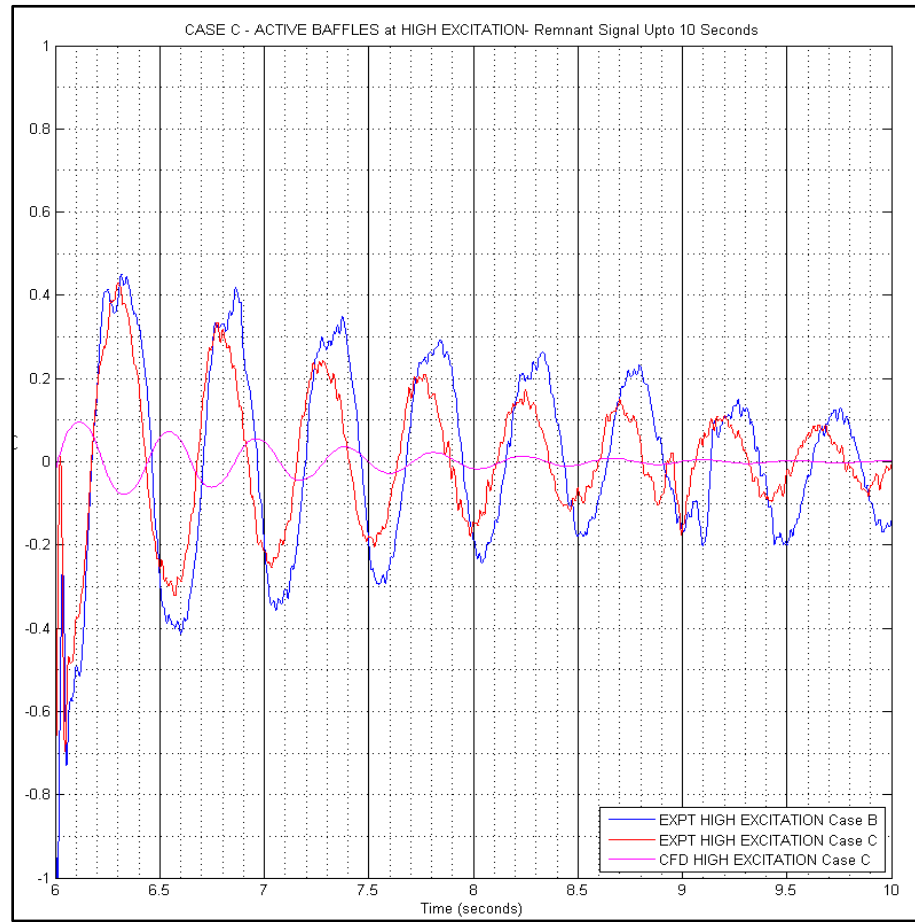


Figure 33: Comparative plot of experimental results from Case B and C, and the CFD results from Case C at high excitation

First by comparing the results between the experimental and CFD cases, it is seen that there are major differences in the magnitude of the peaks. The CFD case assumes pure wall conditions at the micro-baffle regions. This restricts complete motion of the micro-baffles during the slosh and makes the micro-baffle layer more rigid. This brings out a similar pattern for the high excitation case as well. In the experimental cases, the micro-baffles still retain a certain amount of flexibility and thus do not provide a high damping

ratio as compared to the CFD cases. When comparing the two experimental cases, there is a prominent difference in the damping ratio. The activation of the micro-baffles provides a stronger rigid layer over the free surface and dampens the slosh waves.

In general, when observing all the results, a particular difference in the phase shift of the signals is observed. This is where the peak occurrences differ on each trial. This is due to the errors that are produced at the load cells and the variation in the stopping position of the tank after excitation.

To compare the results with a previous PMD method, the active damping method is employed and the damping ratios from their test cases [9].

Table 4: Results from the active damping method

Case Type	Frequency of Excitation (Hz)	Damping Ratio	% Increase in Damping Ratio from free slosh
Free Slosh (Naturally Damped)	2.0	0.0633	n/a
12 Hz, 1 mm amplitude membrane vibration	2.327	0.0955	50.9
12.5 Hz, 1 mm amplitude membrane vibration	2.327	0.1059	67.3

Table 5 and Table 6 show the summary of the results obtained as such and the percentage differences from free slosh.

Table 5: Summary of results for the low excitation condition

	Case Type	Excitation Amplitude (a)	Excitation Frequency (f_a)	Damping Factor (ϵ)	% Increase in Damping Ratio from free slosh
EXPERIMENTAL	A	1.8	2	0.154	n/a
	B	1.8	2	0.195	26.6
	C	1.8	2	0.314	103.9
CFD	A	1.8	2	0.278	n/a
	C	1.8	2	0.615	121.2

Table 6: Summary of results for the high excitation condition

	Case	Excitation Amplitude (a)	Excitation Frequency (f_a)	Damping Factor (ϵ)	% Increase in Damping Ratio from free slosh
EXPERIMENTAL	A	3	2	0.227	n/a
	B	3	2	0.258	13.6
	C	3	2	0.354	55.9
CFD	A	3	2	0.298	n/a
	C	3	2	0.660	121.5

Based on the values of the damping ratios obtained between the active damping results and the floating magnetoactive results. It is observed that the overall values of the damping factors are higher in the active damping method. Despite fact that the active damping experiment utilized a different size cylindrical tank, the results show up to 67% increase in the damping ratio compared to the free slosh condition.

6. CONCLUSIONS AND FUTURE WORK

The initial design and development of the floating magnetoactive micro-baffle is carried out, and the preliminary proof of concept testing of this design as a viable Propellant Management Device (PMD) is conducted.

A preferred model of the micro-baffle is designed and tested for buoyancy and efficiency in grid formation. In addition, the various electromagnets are assembled and tested to provide a fairly strong magnetic field. The most suitable combination of electromagnet is used in the slosh tests and the filtered results are plotted and compared. The computational model is designed based on the assumptions and is tested to simulate the real-life conditions. A favorable trend in the results is achieved for experiments with the magnetoactive micro-baffles by noting that there is a prominent increase in the damping ratio of up to 103%, hence proving the concept of floating magnetoactive micro-baffles as a viable slosh damping mechanism. Comparing this result to an active slosh damping model [9], the micro-baffle method appears to be more effective and provides larger damping values. The CFD results provided a noise-less simulation of the same tests that were conducted. Results vary in magnitude due to the absence of noise from background sources and the need for actuator calibration. This results in the CFD providing extreme ideal conditions for this setup, giving about 120% better damping ratios compared to the free slosh cases.

Though floating magnetoactive micro-baffles offer a significant increase to slosh damping over inactive floating micro-baffles, the scope of the experimentation has been limited. A number of issues need to be addressed and further experimentation must be conducted. Most importantly, the efficiency of floating magnetoactive micro-baffles must be further compared to previously developed PMDs such as diaphragms and wall-fixed baffles. Representation of actual slosh conditions would bolster the effect of the experimental results. At this point in the research, the main challenges are due to the micro-baffle design, magnetic activation source, and the interaction between both elements. Usage of a more magnetically permeable electro active material such as Metglas is to be investigated. Revision to the inner structure geometry and overall shape of the micro-baffle should also be considered. As research progresses, a more consistent method of manufacturing the micro-baffle will be incorporated. Upon refinement of the micro-baffle design, the magnetic activation source will have to be revised. In addition to relating the strength of the magnetic pole to micro-baffle interactions, different magnetic source configurations will be considered with an emphasis of implementing an electromagnet that adjusts to the change in free surface height.

Numerous complications and limitations faced during CFD simulation provide the need for more effective computer resources and research in replicating the magnetoactive micro-baffle setup. A more realistic model is suggested, by neglecting overcoming the assumptions that are initially made, and to simulate this scenario using additional physics models that also need to be considered.

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APPENDICES

Appendix A: Rigid Body Coordinates of the Baffles for DFBI Setup

RIGID BODY COORDINATES

Region	Cx	Cy	Cz
B1	-1.764	-2.333	0
B2	2.903	-0.569	0
B3	-2.631	-0.569	0
B4	-0.803	-0.569	0
B5	1.088	-0.569	0
B6	0.188	-2.333	0
B7	2.015	-2.336	0
B8	0.142	-1.167	0
B9	2.012	-1.167	0
B10	-1.722	-1.167	0
B11	1.112	-2.867	0
B12	-0.788	-2.867	0

Appendix B: Initial Conditions and Field Functions Used

The following settings are used for the initial conditions:

Initial Conditions	
Pressure	
Method	Constant
Dimensions	Pressure
Constant	
Value	0.0 Pa
Turbulence Intensity	
Method	Constant
Dimensions	Dimensionless
Constant	
Value	0.01
Turbulence Specification	
Method	Intensity + Viscosity Ratio
Turbulent Velocity Scale	
Method	Constant
Dimensions	Velocity
Constant	
Value	1.0 m/s
Turbulent Viscosity Ratio	
Method	Constant
Dimensions	Dimensionless
Constant	
Value	10.0
Velocity	
Method	Constant
Dimensions	Velocity
Coordinate System	Laboratory
Constant	
Value	[0.0, 0.0, 0.0] m/s
Volume Fraction	
Method	Composite
Dimensions	Dimensionless
Composite	
Water	
Method	Field Function
Dimensions	Dimensionless
Field Function	
Scalar Function	Fluid_InitiaVF
Air	
Method	Field Function
Dimensions	Dimensionless
Field Function	
Scalar Function	Air_InitiaVF

Figure 34: Initial conditions settings

The following field functions are used for the Translational motion to generate the lateral excitation by Simple Harmonic Motion (SHM) based on Equation (6).

Air_InitialVF	
Function Name	UserFieldFunction_1
Type	Scalar
Dimensions	Dimensionless
Definition	$(\$Centroid[2] > 0) ? 1:0$
Inverse Distance Weight	☐
Assembly code	$(if (> \$Centroid[2] 0) 1 0)$
Ignore Boundary Values	☐
Fluid_InitialVF	
Function Name	UserFieldFunction_1_2
Type	Scalar
Dimensions	Dimensionless
Definition	$(\$Centroid[2] < 0) ? 1:0$
Inverse Distance Weight	☐
Assembly code	$(if (< \$Centroid[2] 0) 1 0)$
Ignore Boundary Values	☐

Figure 35: Air and fluid initial VOF settings

 high_wall_x	
Function Name	highamp
Type	Vector
Dimensions	Velocity
Definition	interpolateTable(@Table("highampwall"), "time", LINEAR, "vel_x", "vel_y", "vel_z", \$Time)
Inverse Distance Weight	☐
Assembly code	interpolateTable(@Table("highampwall"), "time", LINEAR, "vel_x", "vel_y", "vel_z", \${Time})
Ignore Boundary Values	☐
 low_wall_x	
Function Name	lowamp
Type	Vector
Dimensions	Velocity
Definition	interpolateTable(@Table("lowampwall"), "time", LINEAR, "vel_x", "vel_y", "vel_z", \$Time)
Inverse Distance Weight	☐
Assembly code	interpolateTable(@Table("lowampwall"), "time", LINEAR, "vel_x", "vel_y", "vel_z", \${Time})
Ignore Boundary Values	☐

Figure 36: Settings to read a table containing SHM data for tank motion

	time	vel_x	vel_y	vel_z	amp	freq	pi
0	0	0	0	0	0.003	2	3.1416
1	0.1	0.0116	0	0	0.003	2	3.1416
2	0.2	-0.0305	0	0	0.003	2	3.1416
3	0.3	-0.0305	0	0	0.003	2	3.1416
4	0.4	0.0116	0	0	0.003	2	3.1416
5	0.5	0.0377	0	0	0.003	2	3.1416
6	0.6	0.0116	0	0	0.003	2	3.1416
7	0.7	-0.0305	0	0	0.003	2	3.1416
8	0.8	-0.0305	0	0	0.003	2	3.1416
9	0.9	0.0116	0	0	0.003	2	3.1416
10	1	0.0377	0	0	0.003	2	3.1416
11	1.1	0.0117	0	0	0.003	2	3.1416
12	1.2	-0.0305	0	0	0.003	2	3.1416

Figure 37: Sample Table for importing motion data into the field functions to generate
SHM

Appendix C: Determination of the Damping Coefficient

The Following sample calculation shows a method to determine the damping ratio of the remnant slosh waves based on the data peak analysis.

First considering two peaks at different times t_1 and t_2 , the corresponding amplitude values y_1 and y_2 :

$y_1 = 0.002963$ at $t_1 = 8.28$ seconds, and $y_2 = 0.002250$ at $t_2 = 8.84$ seconds

The logarithmic decrement between these two peaks is given as:

$$\delta = \ln \left(\frac{x_1}{x_2} \right)$$

$$\therefore \delta = \ln \left(\frac{0.002963}{0.002250} \right)$$

$$\therefore \delta = 0.274872$$

$$\delta = \frac{2\pi\varepsilon}{\sqrt{(1-\varepsilon^2)}}$$

Rearranging we get:

$$\varepsilon = \frac{\delta}{\sqrt{(2\pi + \delta^2)}}$$

ε is the damping ratio. Solving this equation for ε , gives the Damping Ratio,

$$\varepsilon = 0.043704$$